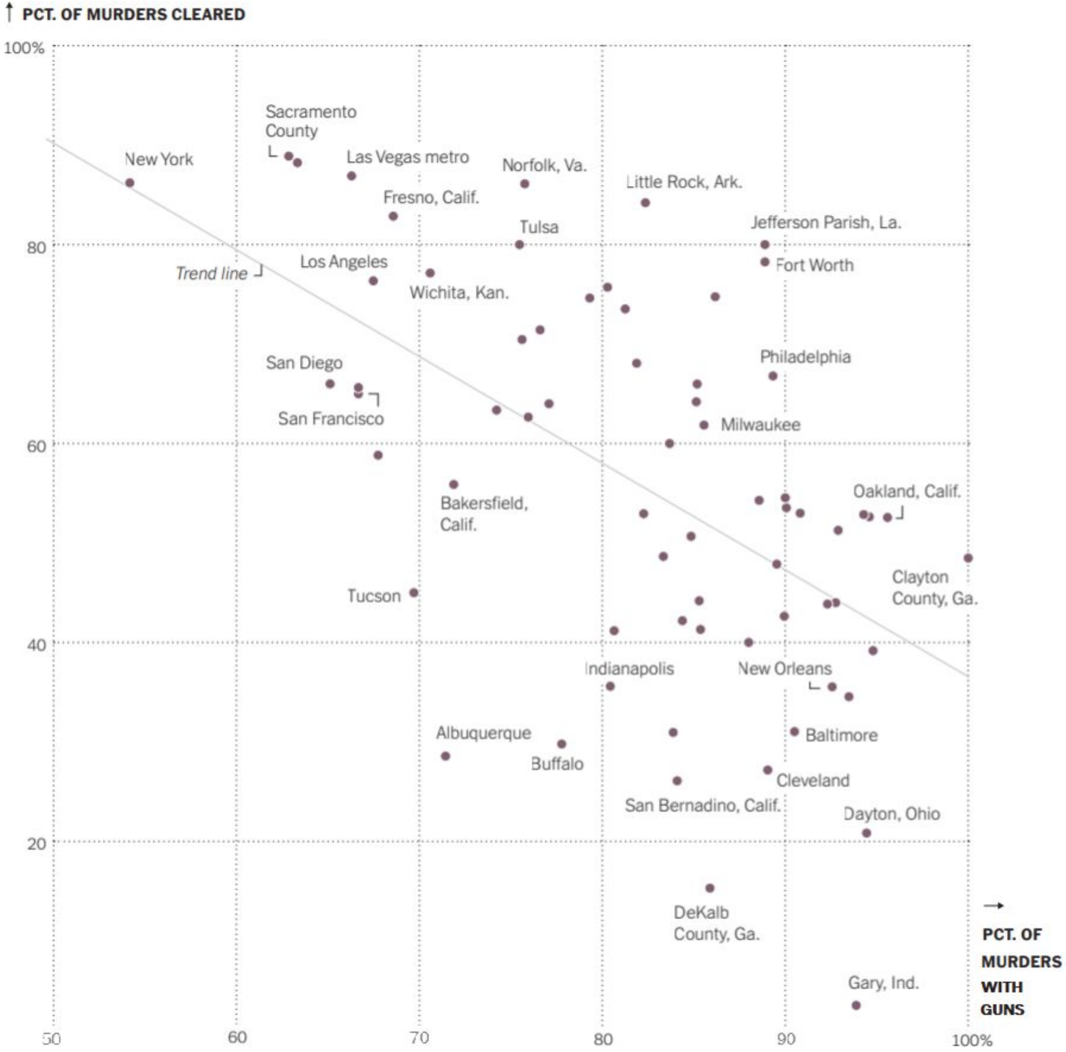
Murders in the U.S.

MATH 213

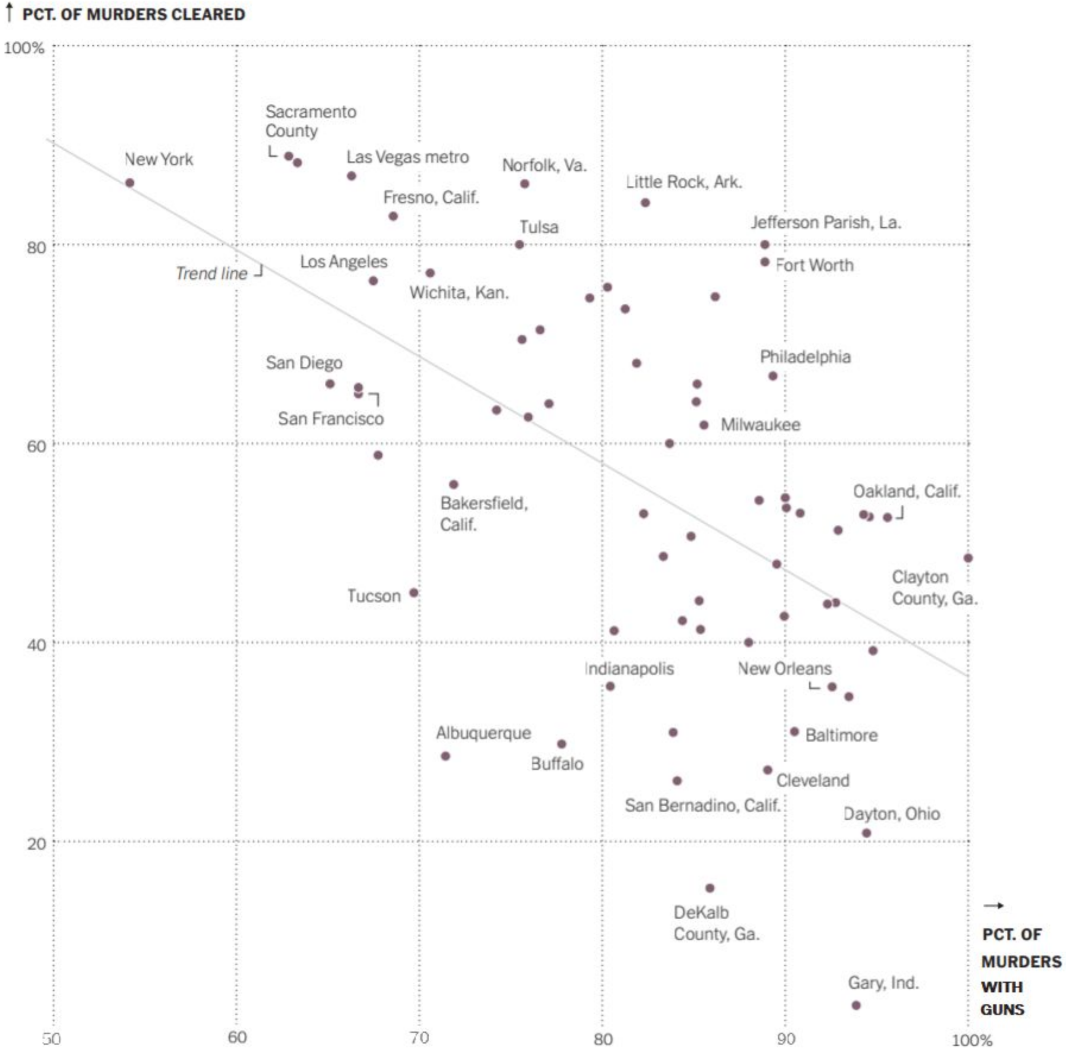


Group 1

This graph is from the NYTimes. Please don't search for the article just yet. Here is what the opening of the article says about the graph:

"Is the likelihood of solving a murder case less if the weapon is a gun?

This graph shows the relationship between the percentage of murders committed with guns for the 67 cities that had 30 or more murders in 2019 and the percentage of all murders solved."



1. Tell me what you can about Norfolk, VA. Be specific.

Norfolk has about 86% of their murders solved however only about 60% of those are with a gun.

2. Tell me what you can about New York, NY. Be specific.

New york has about 86% of their murders solved however only about 54% of those are with a gun

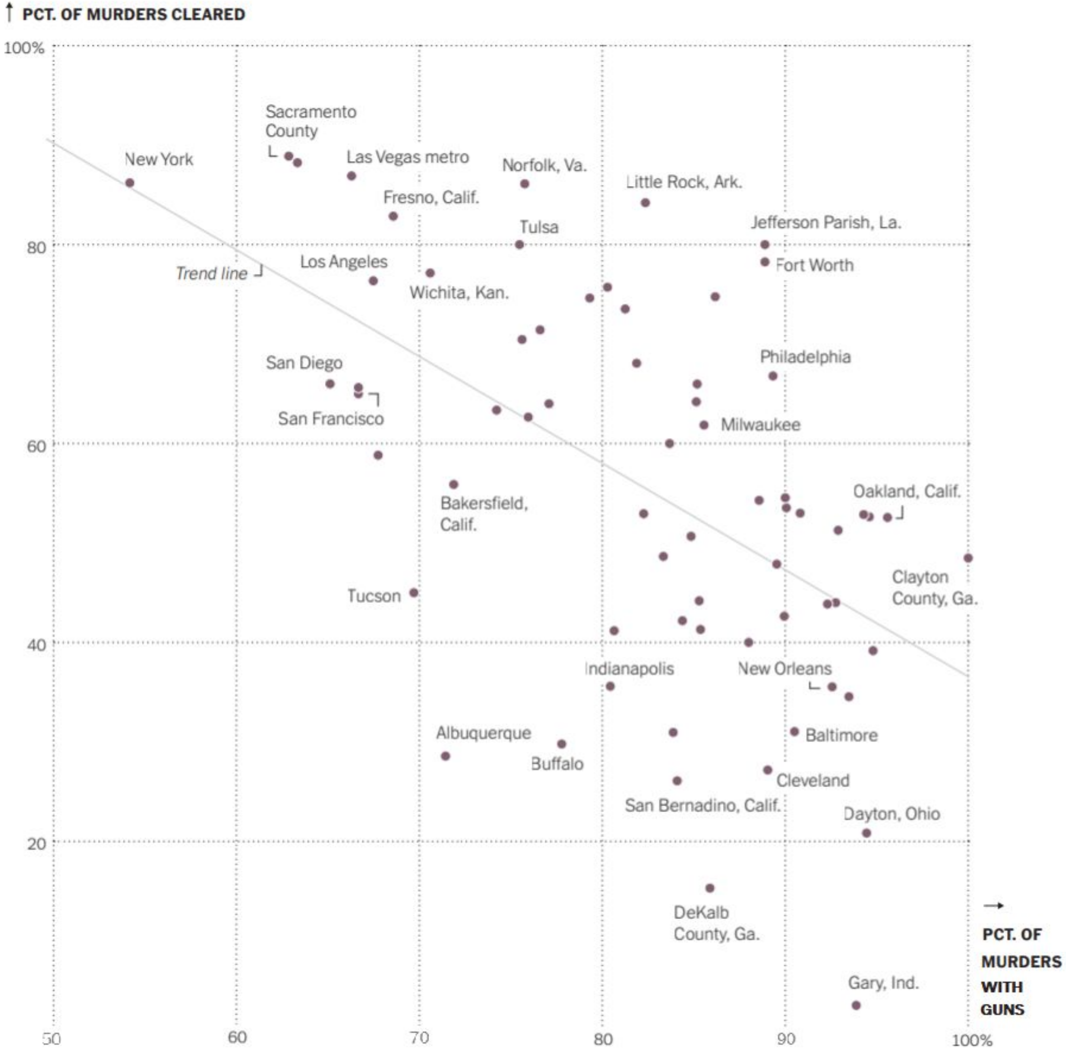
3. Fill in the blank:

Generally speaking, for the cities represented in this graph, the greater the percentage of murders committed using a gun, the **smaller** the percentage of murder cases that are solved.

4. What is the **sample** for this visualization?

Cities that had over 30 or more gun murders in 2019

5. What is the **population of interest** to the person who created this visualization, would you say?



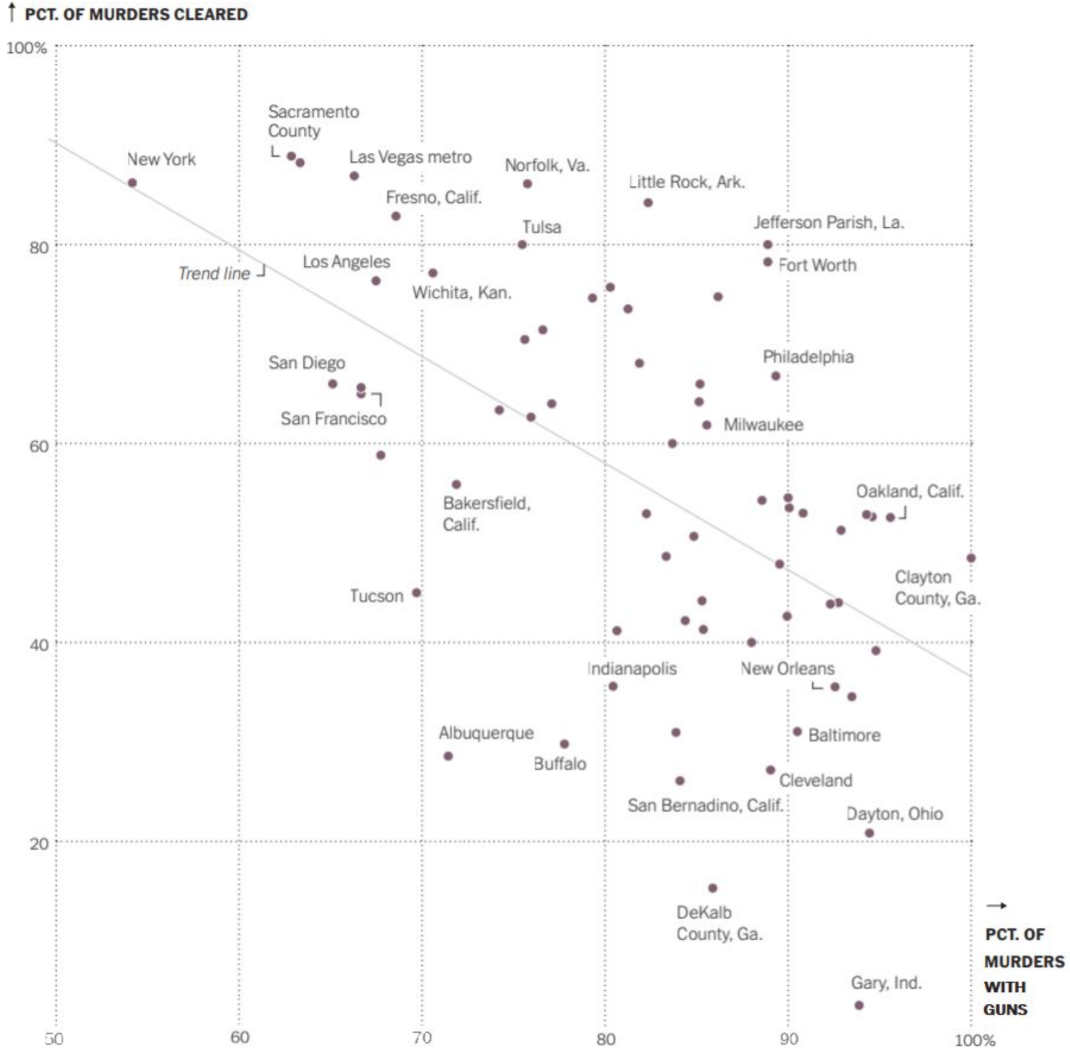
First, pretend the cases are "the murders".

6. What **variables (for cases = *murders*)** were used to create this visualization? Be careful with wording. You should have at least 3 variables.

murder with gun, murder cleared, City

7. Since each point on the graph represents a city, the cases here are actually "the cities." What **variables are recorded about each *city*** in this visualization?

Number of murders with a gun, number of murders cleared.

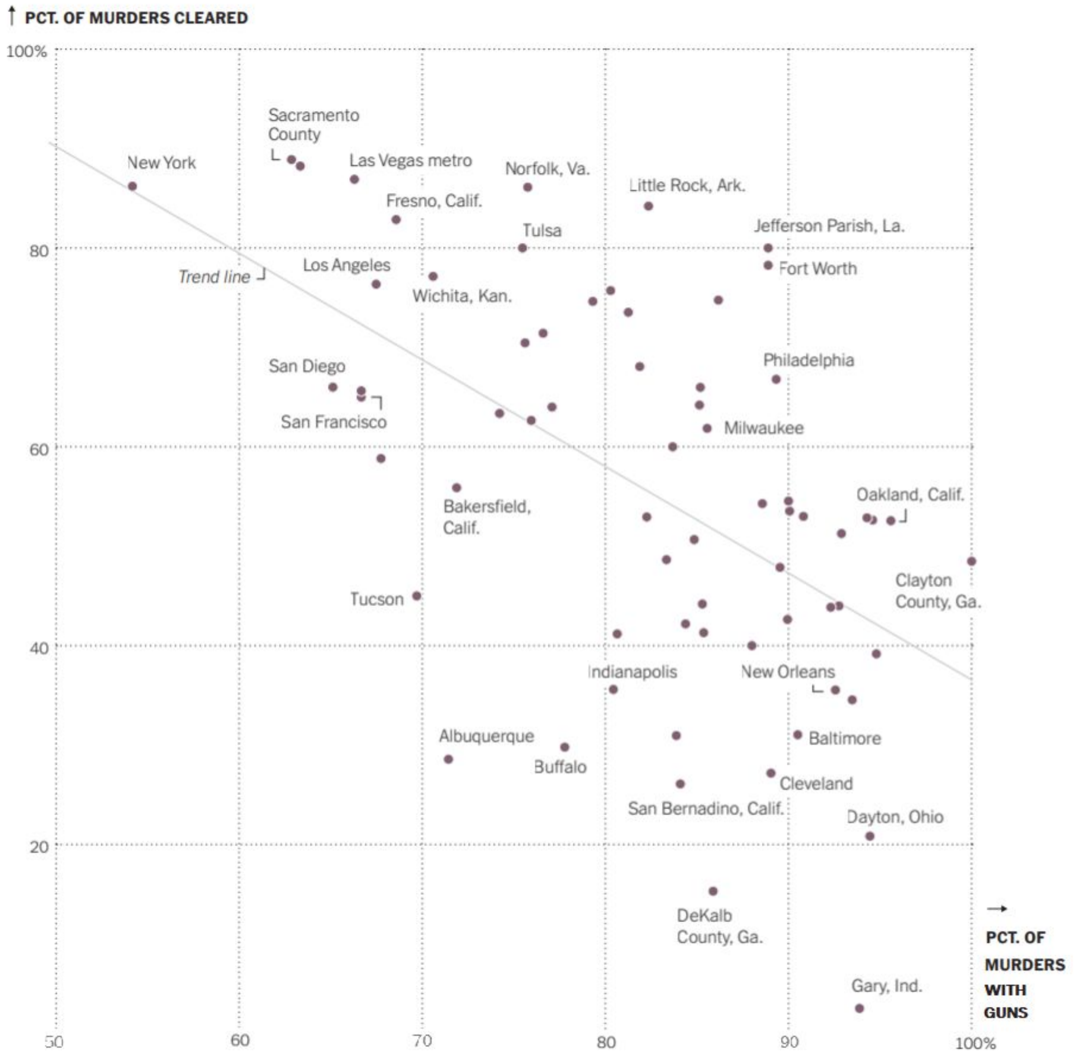


8. Can we draw the conclusion that the more murders there are committed using guns *causes* fewer murder cases to be solved? Discuss.

We can draw the conclusion that the use of guns in a murder, decreases the probability that the murder will be solved. We see this in the graph due to the trendline of murders cleared going down as murders with guns increase.

9. Can we generalize any conclusions we draw from this graph to all cities in the U.S.? Discuss.

We likely cannot generalize any definitive conclusions about any city based on this data, though there is a slight trend in the percent of murders solved and the percent that were committed with guns it likely cannot be generalized to any specific city because very few cities fall on or near the trend line in this graph.



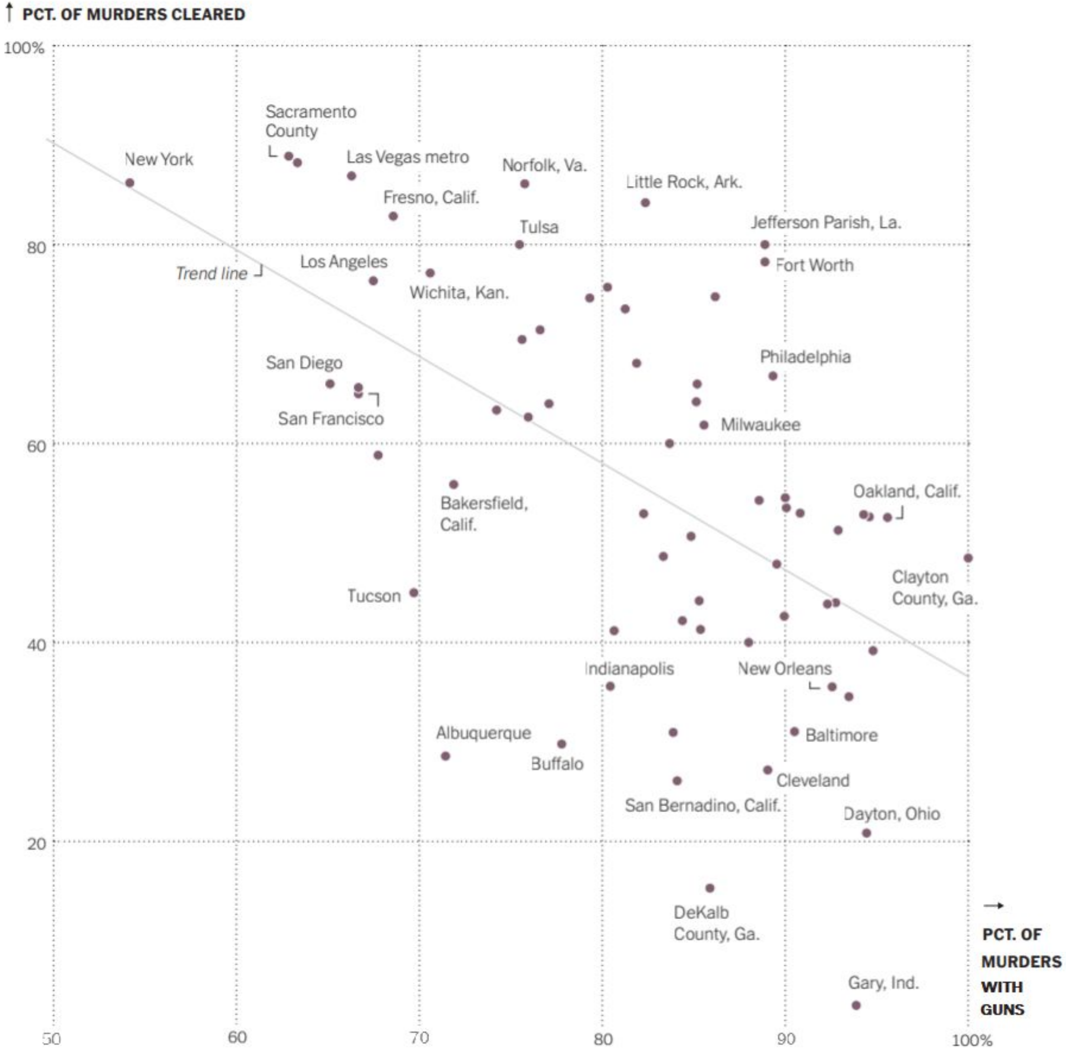
10. Propose a possible confounding variable that might explain the association between percentage of murders with guns and percentage of murders cleared.

Murders that use more than one murder weapon, including a gun.

11. Link between your confounding variable and the percentage of murders committed with guns:

12. Link between your confounding variable and the percentage of murders that are solved:

Murders that use a gun and another weapon have a higher chance to be solved because there is double the evidence for investigators to find in terms of murder weapon.



Group 2

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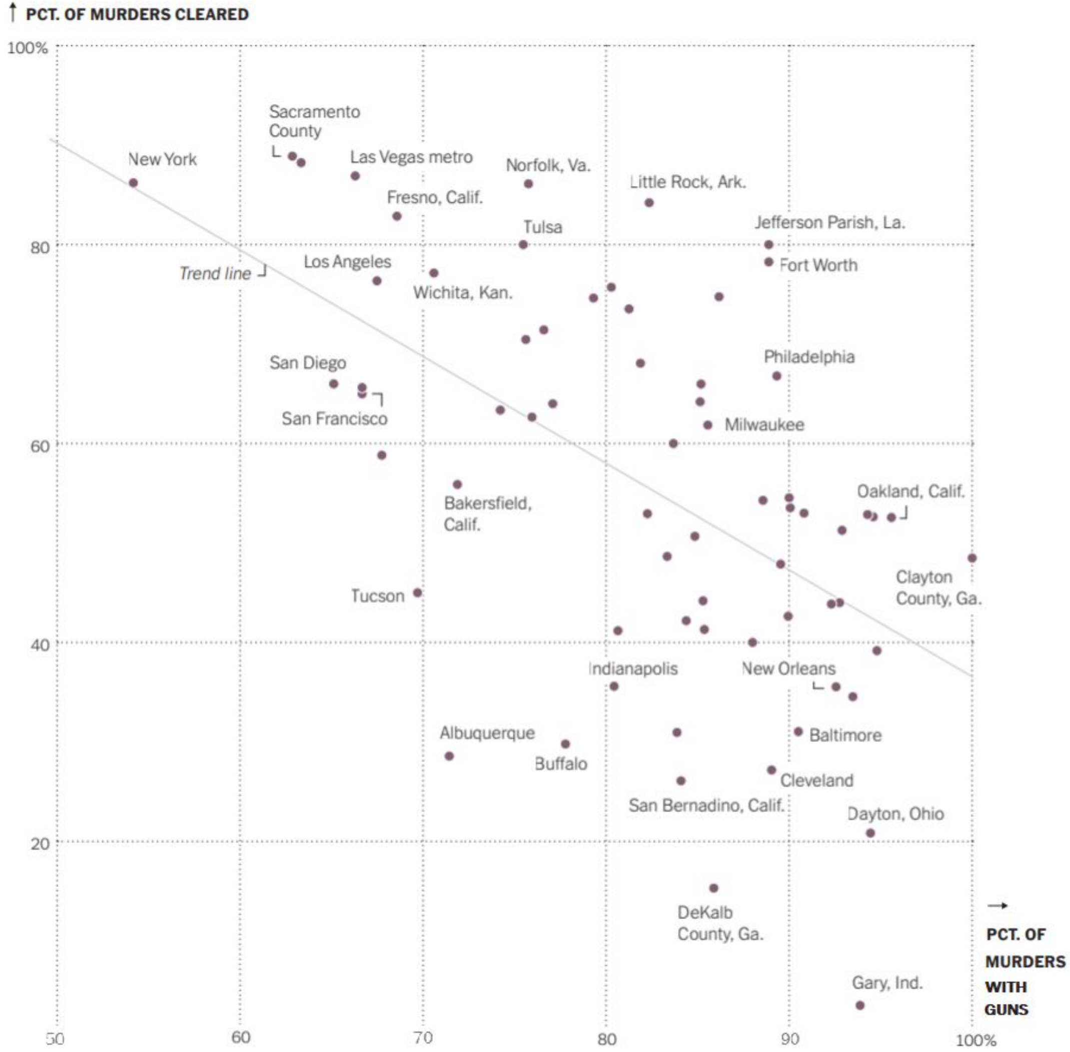


1. Tell me what you can about Norfolk, VA. Be specific.

Norfolk has one of the highest percentage of murders cleared in the dataset at about 86%. About 75% of murders were with guns in Norfolk.

2. Tell me what you can about New York, NY. Be specific.

New York has a high percentage of murders cleared with ~85% cleared and about 55% were with guns.

3. Fill in the blank:

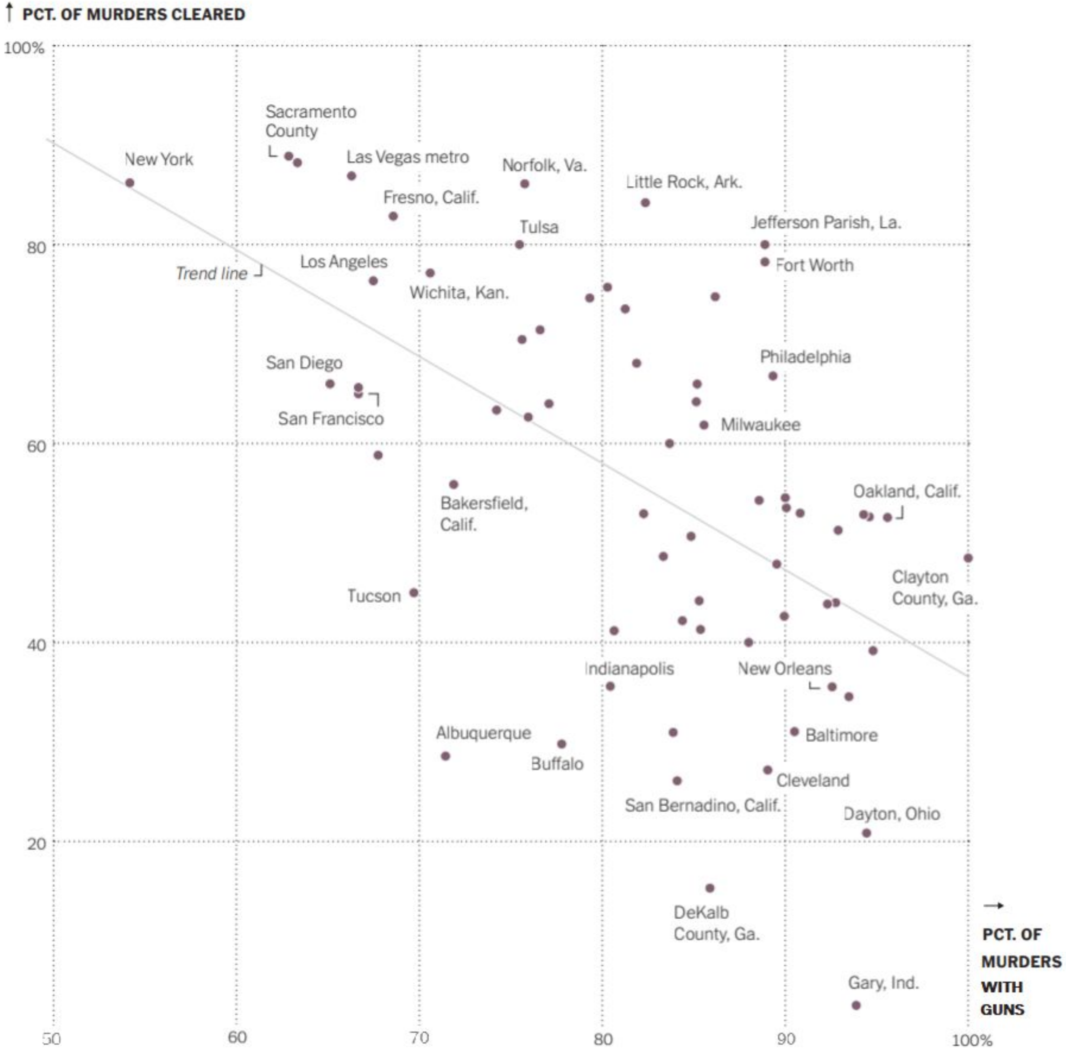
Generally speaking, for the cities represented in this graph, the greater the percentage of murders committed using a gun, the smaller the percentage of murder cases that are solved.

4. What is the **sample** for this visualization?

The sample is the 67 cities in 2019 that had 30 or more murders in the United States.

5. What is the **population of interest** to the person who created this visualization, would you say?

It would likely be all the cities in the United States.



First, pretend the cases are “the murders”.

6. What **variables (for cases = *murders*)** were used to create this visualization? Be careful with wording. You should have at least 3 variables.

Type of murder (shooting, stabbing, etc)

Location

Cleared or not

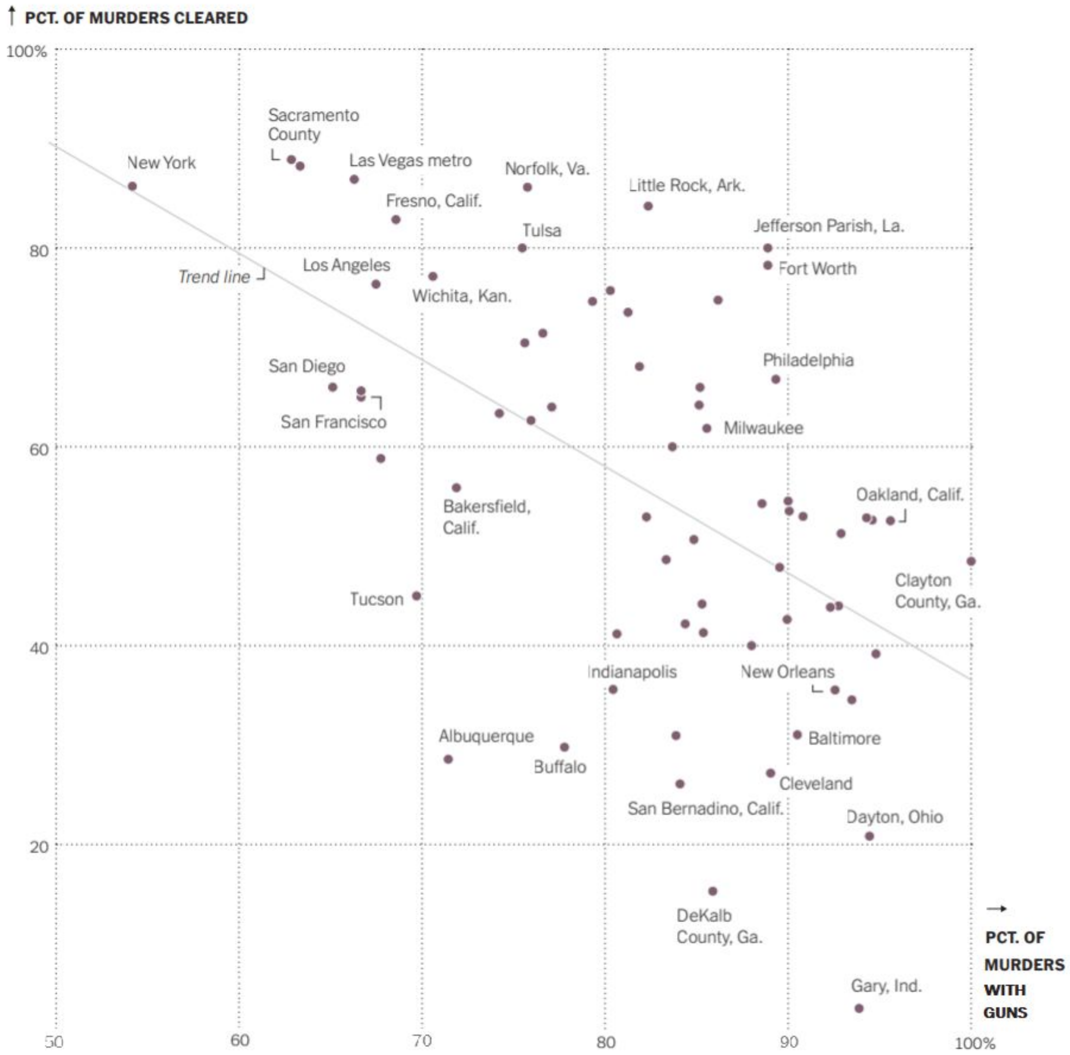
7. Since each point on the graph represents a city, the cases here are actually “the cities.” What **variables are recorded about each *city*** in this visualization?

Location

Number of murders

Percent of murders with guns

Percent of murders cleared

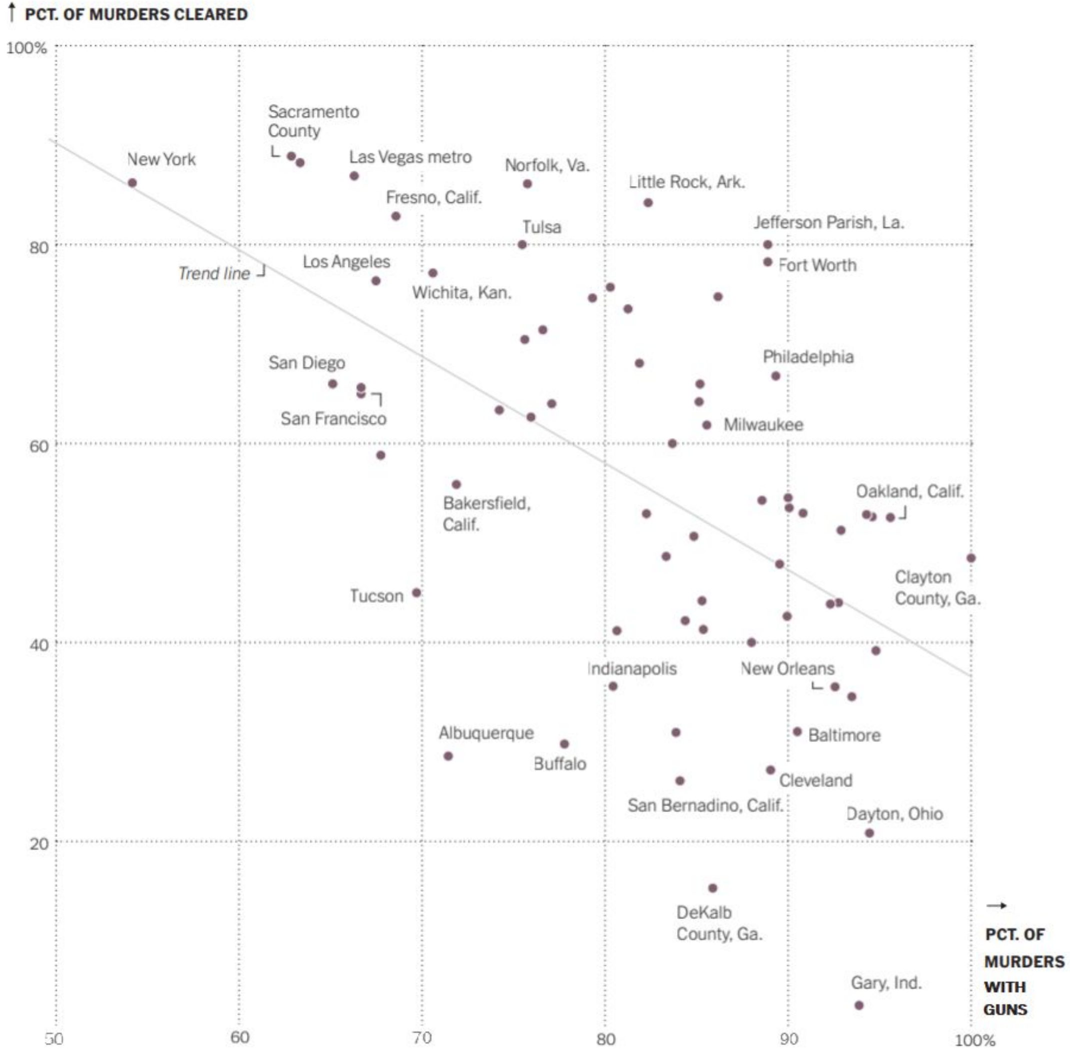


8. Can we draw the conclusion that the more murders there are committed using guns *causes* fewer murder cases to be solved? Discuss.

No, since the study is observational and can not prove causation but association. Also, no variables were manipulated.

9. Can we generalize any conclusions we draw from this graph to all cities in the U.S.? Discuss.

Yes, the study includes cities from all over the United States and take into account the population of interest. However, cities with high murder rates were only selected.



10. Propose a possible confounding variable that might explain the association between percentage of murders with guns and percentage of murders cleared.

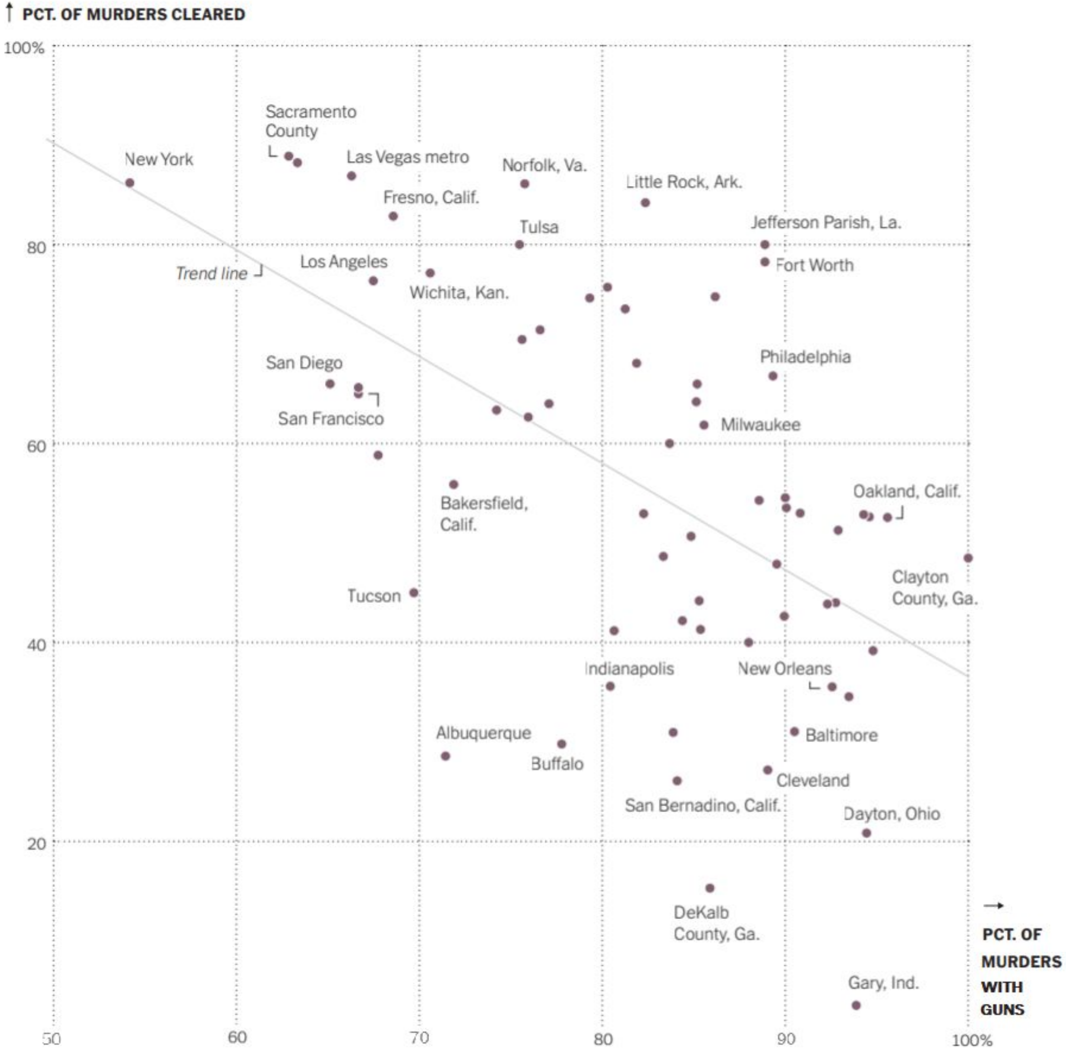
Lack of police force

11. Link between your confounding variable and the percentage of murders committed with guns:

Lack of police force causes more murders and more illegal guns to be on the street.

12. Link between your confounding variable and the percentage of murders that are solved:

Lack of police leads to authorities being focused on maintaining peace rather than investigating previous murders



Group 3

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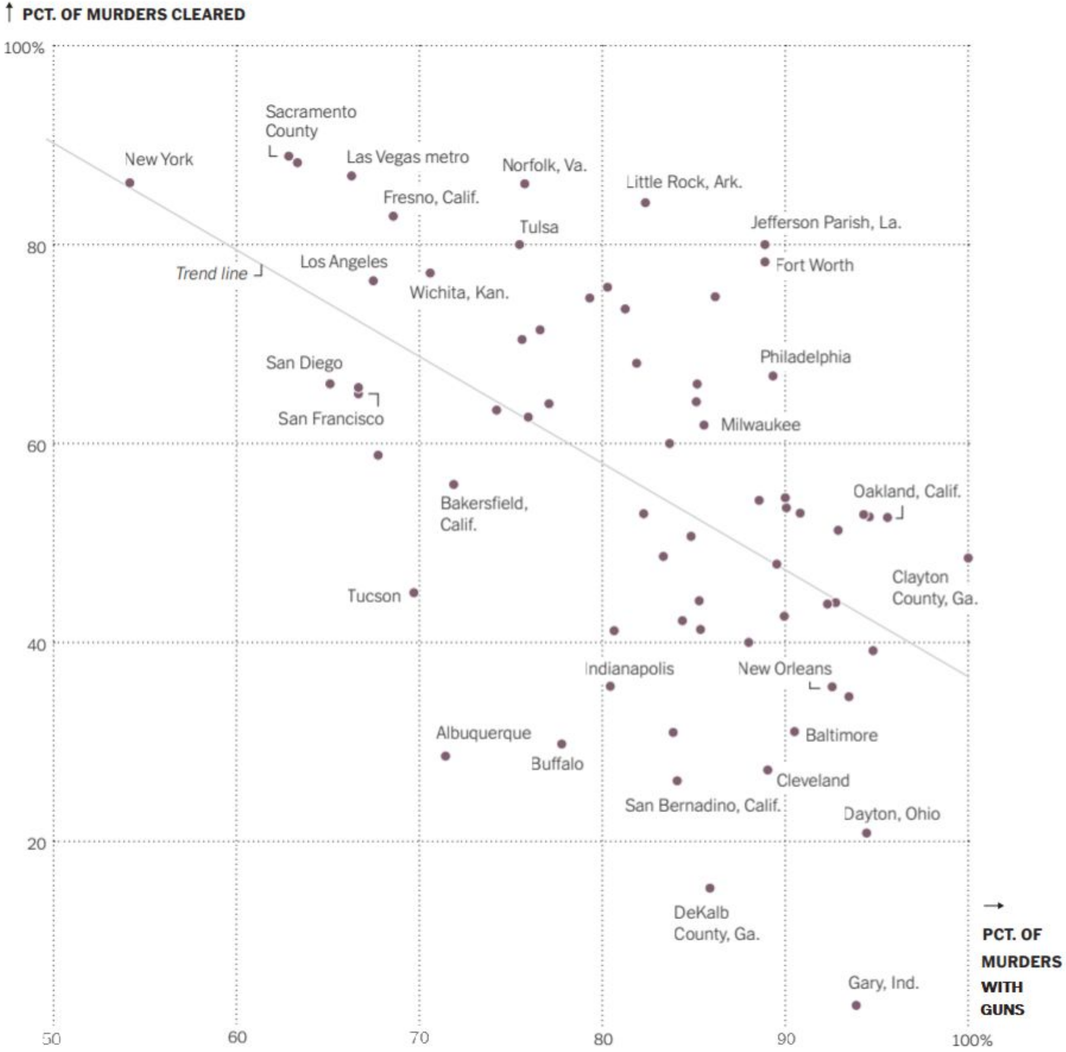


1. Tell me what you can about Norfolk, VA. Be specific.

Norfolk, Va had about 75 percent of murders with guns, with about 90 percent of those murders were cleared. Which is a high level of murders cleared.

2. Tell me what you can about New York, NY. Be specific.

New York had 55 percent for murderers with guns, and again had about 90 percent of those murders clearned.

3. Fill in the blank:

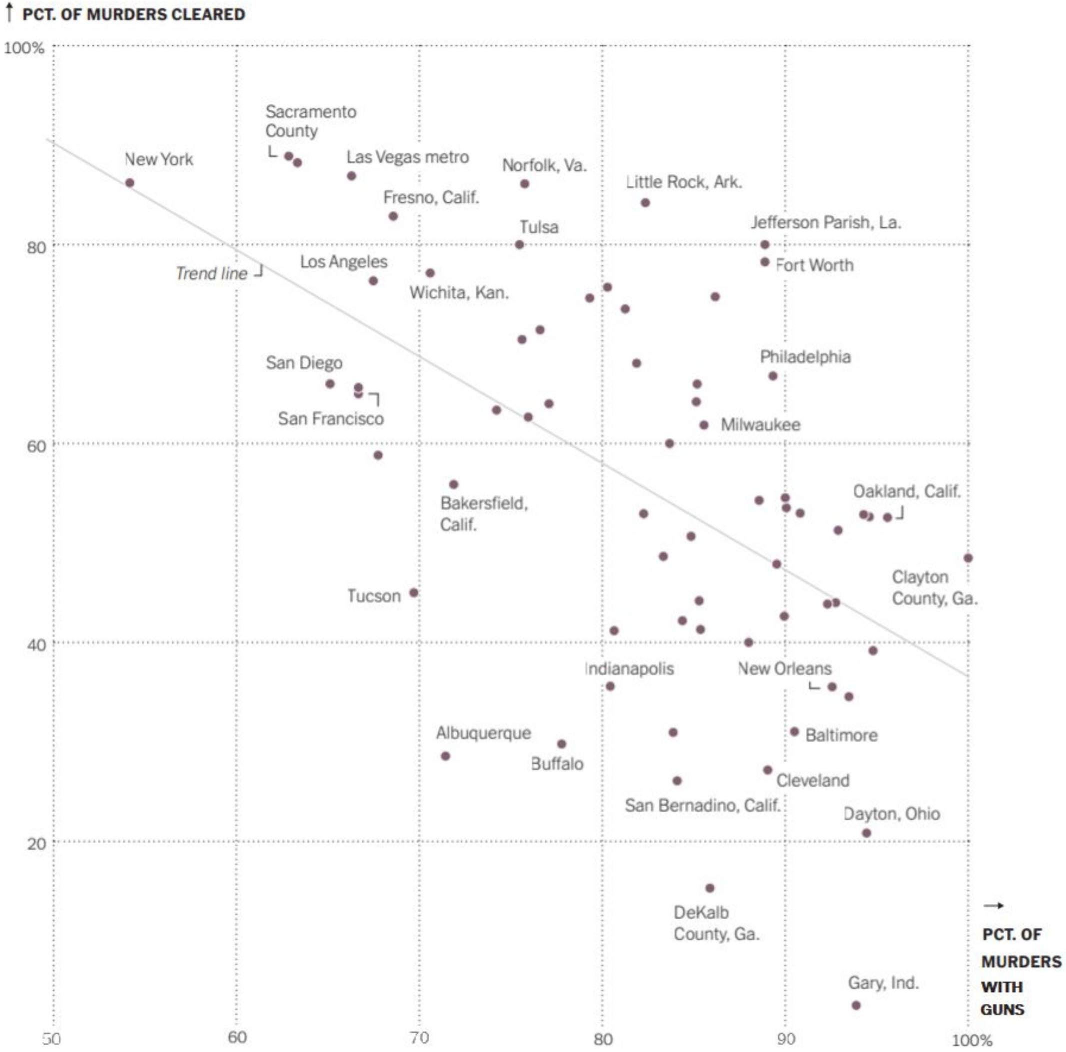
Generally speaking, for the cities represented in this graph, the greater the percentage of murders committed using a gun, the higher the percentage of murder cases that are solved.

4. What is the **sample** for this visualization?

Murder cases in 67 major cities.

5. What is the **population of interest** to the person who created this visualization, would you say?

All the cities in the US.



First, pretend the cases are “the murders”.

6. What **variables (for cases = *murders*)** were used to create this visualization? Be careful with wording. You should have at least 3 variables.

   Did the murderer use a gun

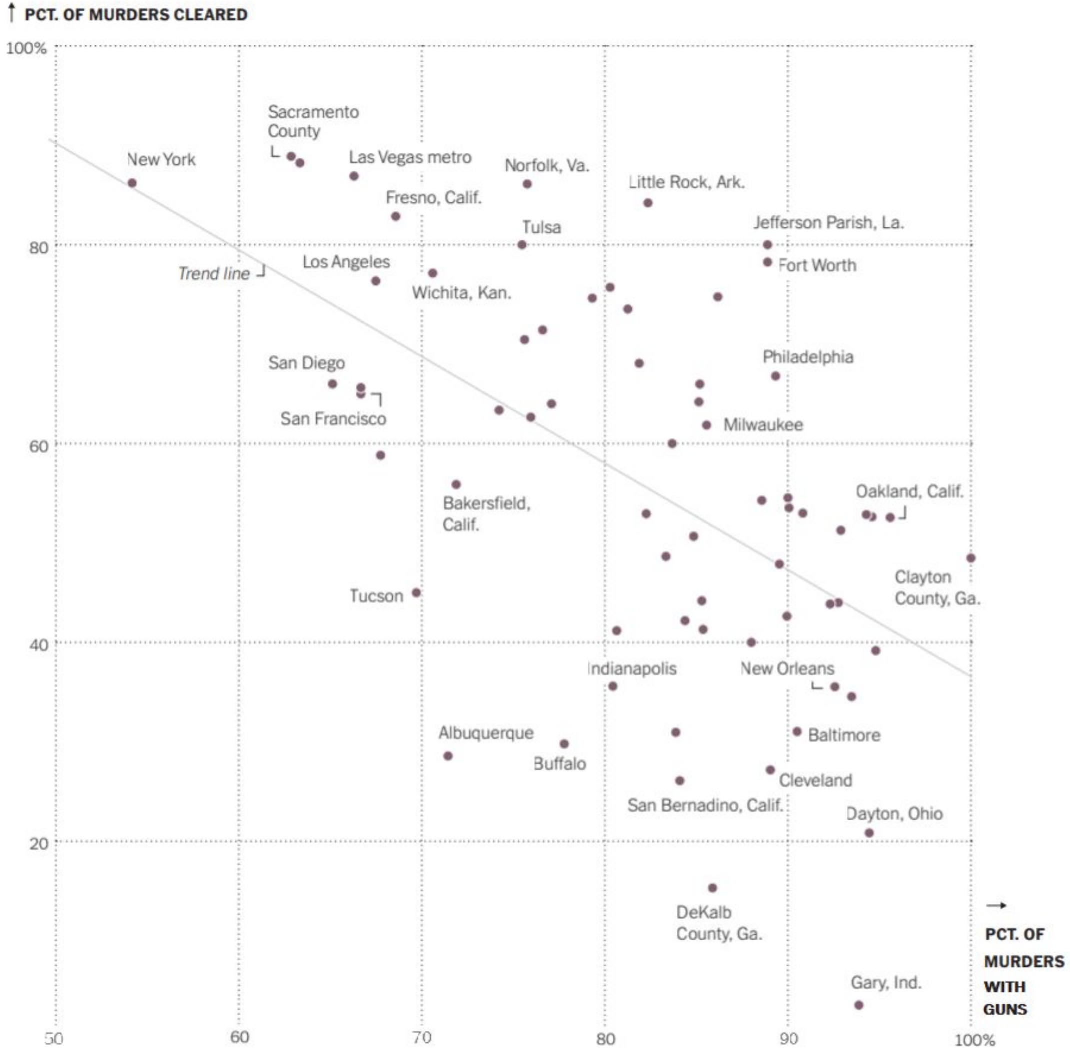
   Was the murderer clearned

   Where did the murder take place

7. Since each point on the graph represents a city, the cases here are actually “the cities.” What **variables are recorded about each *city*** in this visualization?

   Percent of murderers cleared.

   Percent of murderers with guns

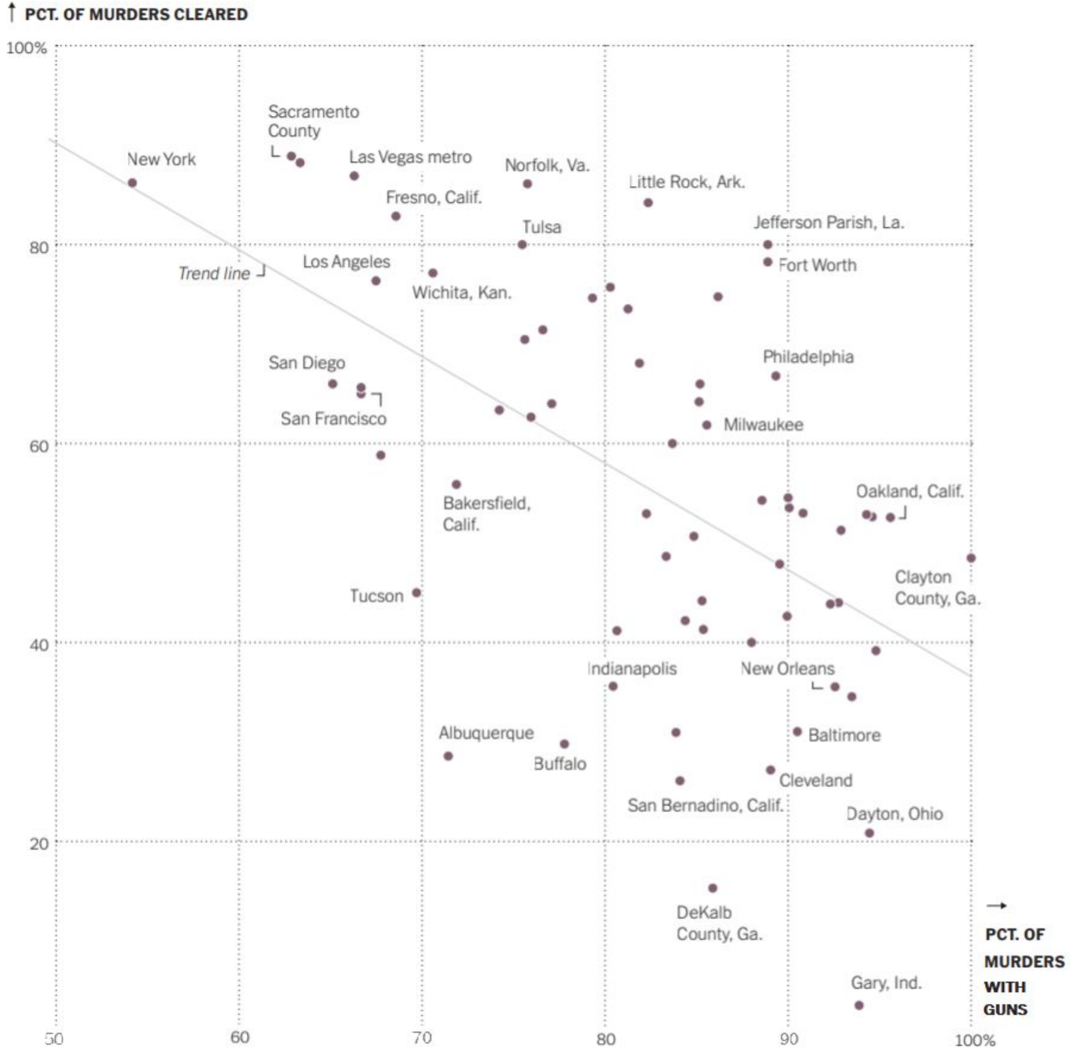


8. Can we draw the conclusion that the more murders there are committed using guns *causes* fewer murder cases to be solved? Discuss.

Since the variables of this study aren't being manipulated, we can't draw a causal conclusion from the study.

9. Can we generalize any conclusions we draw from this graph to all cities in the U.S.? Discuss.

No we cannot, the study was not randomly selected, and instead was selected if they had 30 or more murders.



10. Propose a possible confounding variable that might explain the association between percentage of murders with guns and percentage of murders cleared.

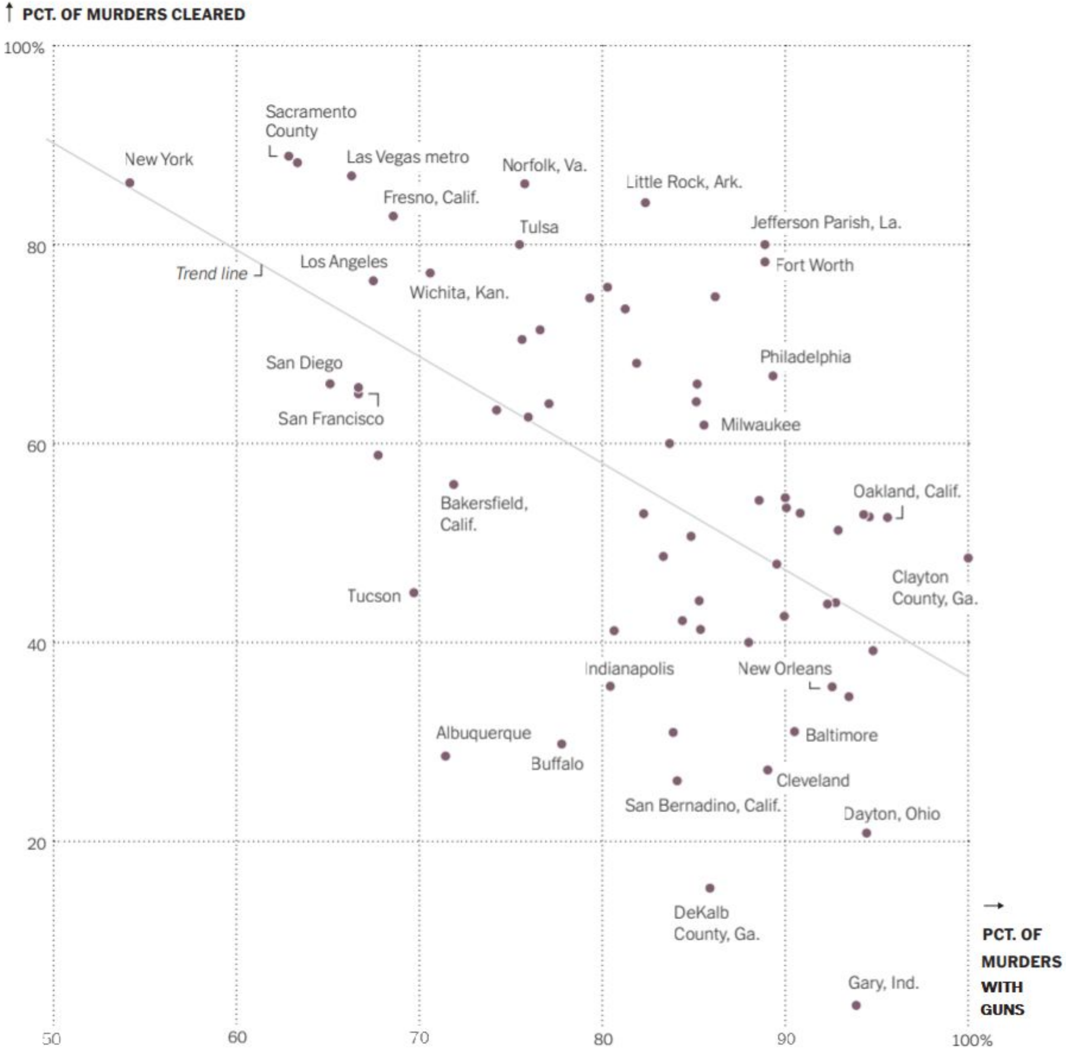
Amount of illegal guns in the city.

11. Link between your confounding variable and the percentage of murders committed with guns:

If you have more illegal guns in your city there will be more murders being committed with those guns.

12. Link between your confounding variable and the percentage of murders that are solved:

Also illegal guns are harder to track, so it could explain the less percent of murders cleared, since legal guns generally have lots of restrictions.



Group 4

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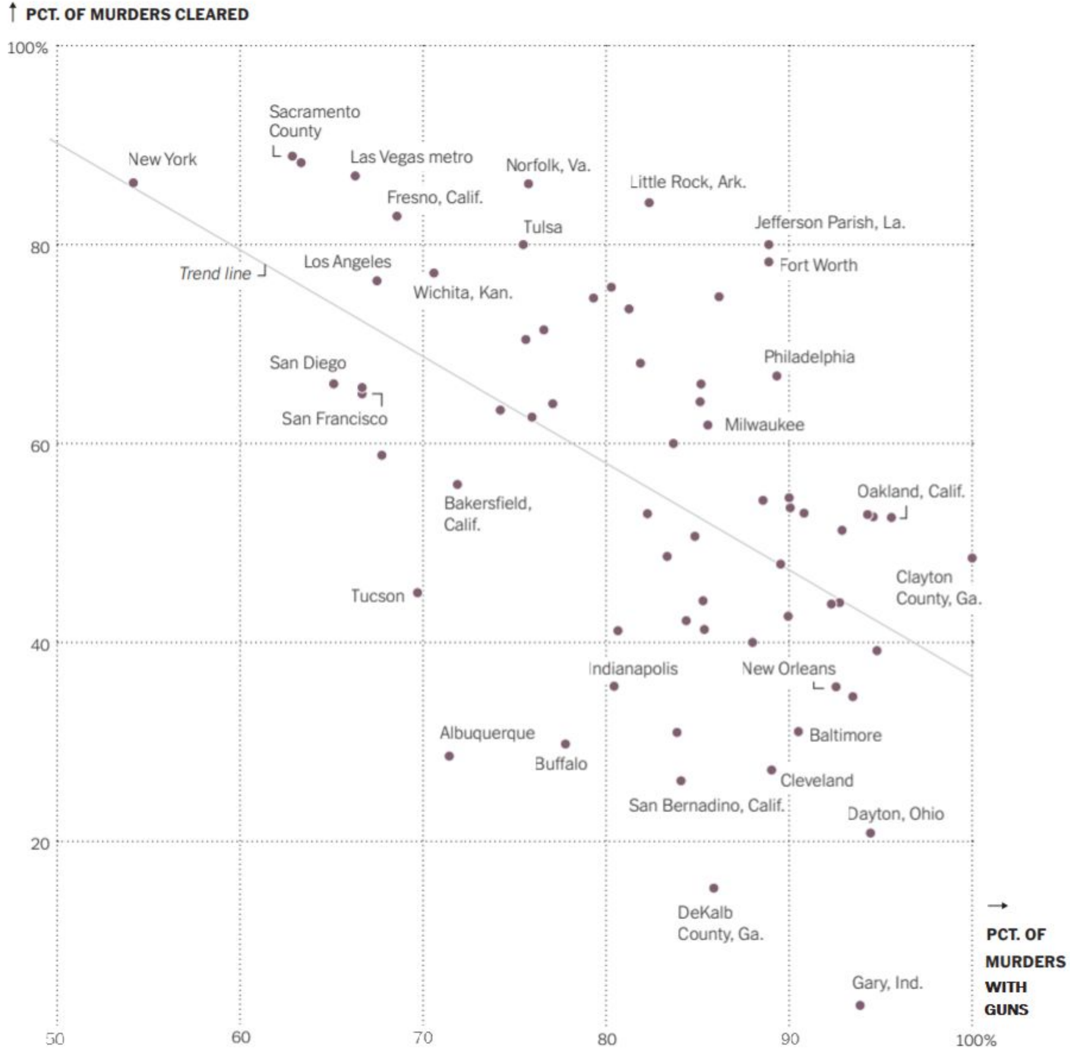


1. Tell me what you can about Norfolk, VA. Be specific.

In Norfolk, VA, around 75% of murders were with guns, and around 85% of murders were cleared.

2. Tell me what you can about New York, NY. Be specific.

   In New York, NY, around 55% of murders were with guns, and around 85% of murders were cleared.

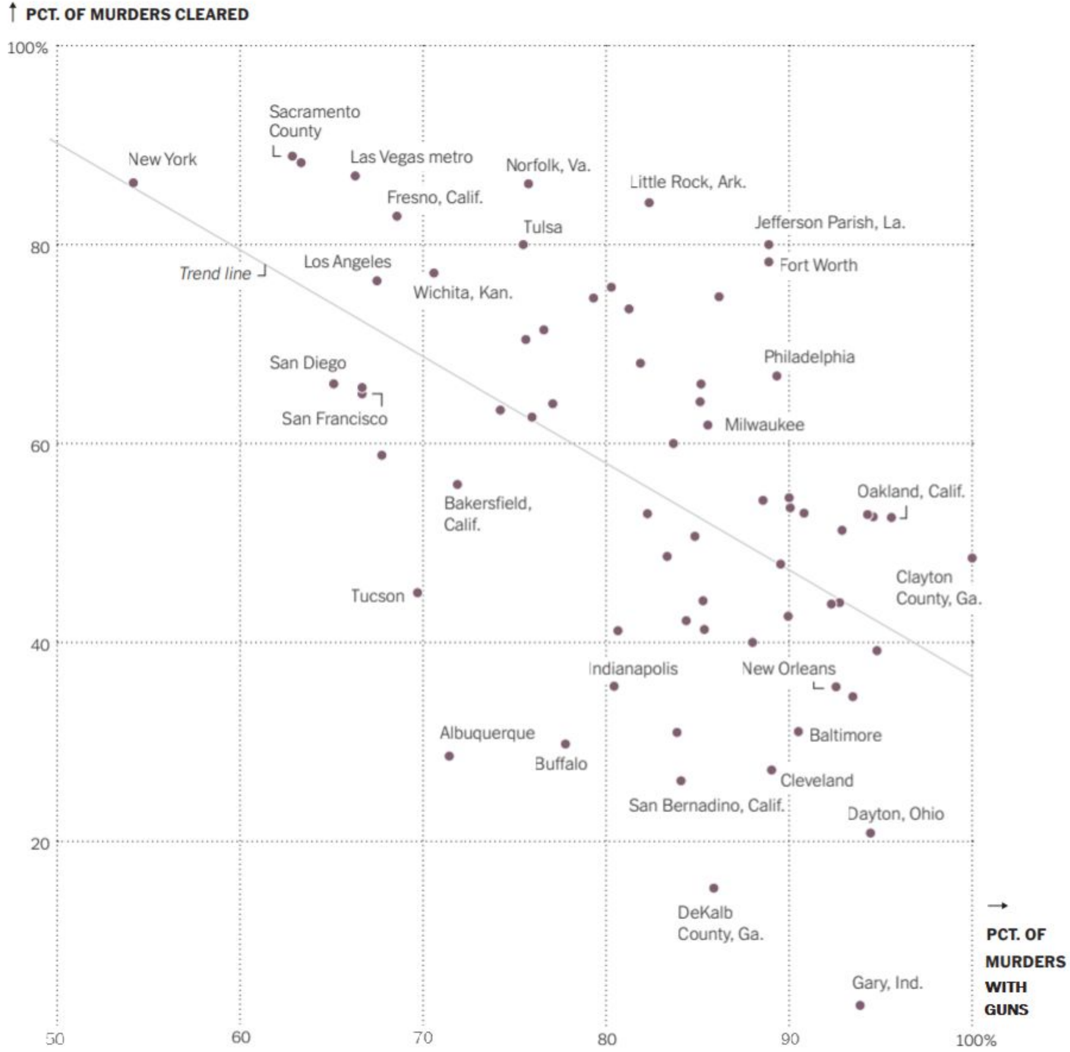
3. Fill in the blank:

Generally speaking, for the cities represented in this graph, the greater the percentage of murders committed using a gun, the **LOWER** the percentage of murder cases that are solved.

4. What is the **sample** for this visualization?

Murders in American cities with over 30 murders in 2019

5. What is the **population of interest** to the person who created this visualization, would you say?
   American cities



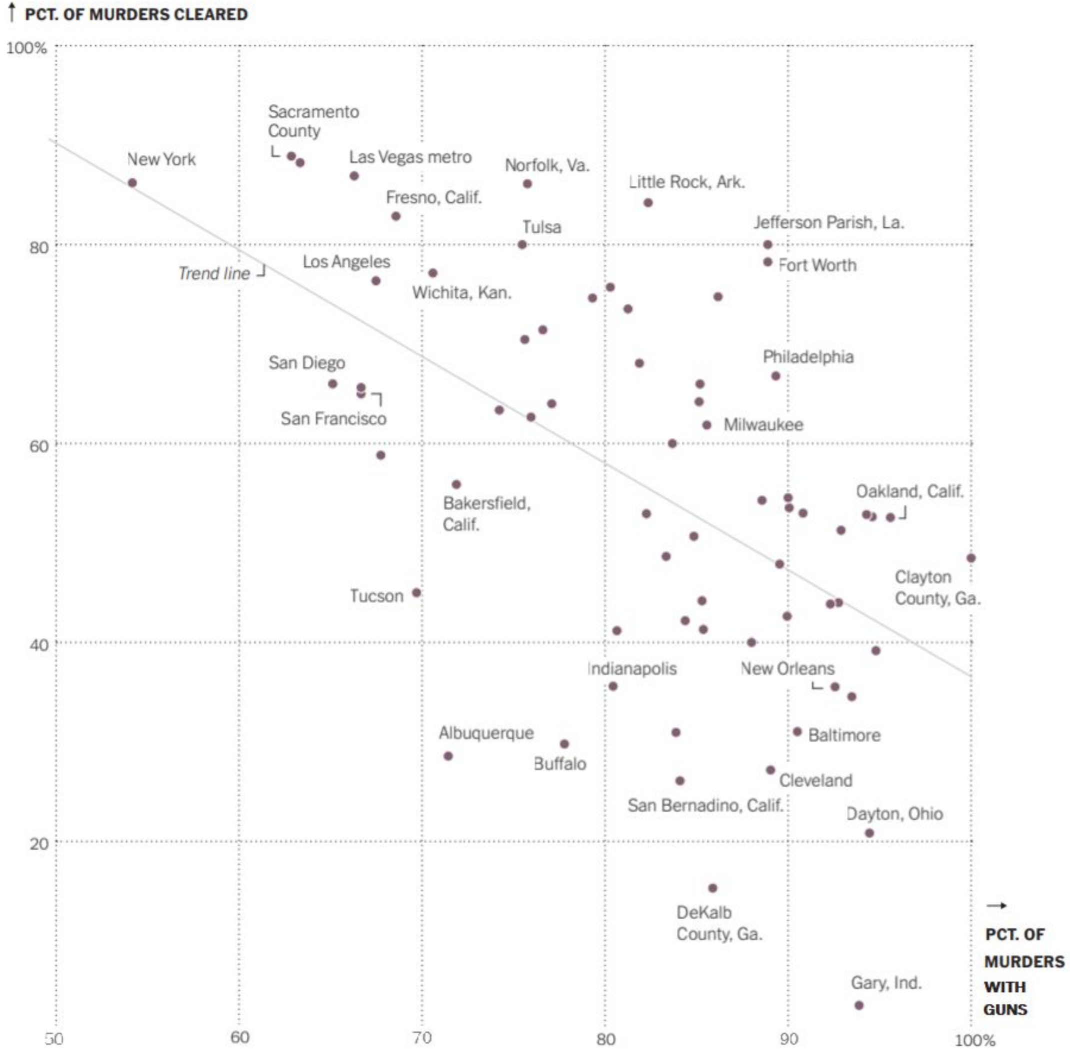
First, pretend the cases are “the murders”.

6. What **variables (for cases = *murders*)** were used to create this visualization? Be careful with wording. You should have at least 3 variables.

- Weapon of Murder(“gun”, “knife”, etc.)
- City of Murder
- Cleared(Y/N)

7. Since each point on the graph represents a city, the cases here are actually “the cities.” What **variables are recorded about each *city*** in this visualization?

- Number of Murders Cleared
- Number of Murders with a Gun

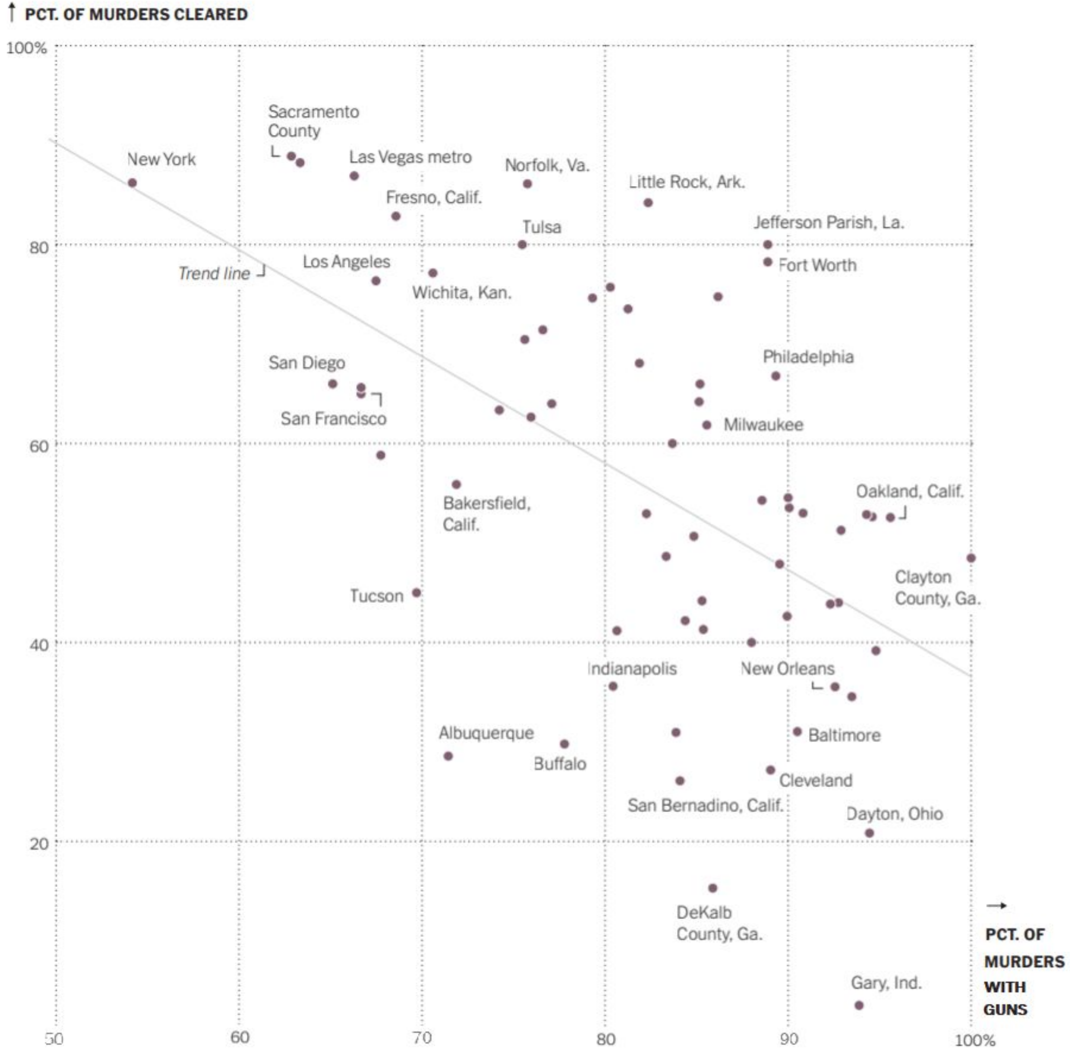


8. Can we draw the conclusion that the more murders there are committed using guns *causes* fewer murder cases to be solved? Discuss.

Because this is an observational study, we cannot declare causation.

9. Can we generalize any conclusions we draw from this graph to all cities in the U.S.? Discuss.

No, because this graph only focuses on big cities which isn't a representation of all cities in America



10. Propose a possible confounding variable that might explain the association between percentage of murders with guns and percentage of murders cleared.

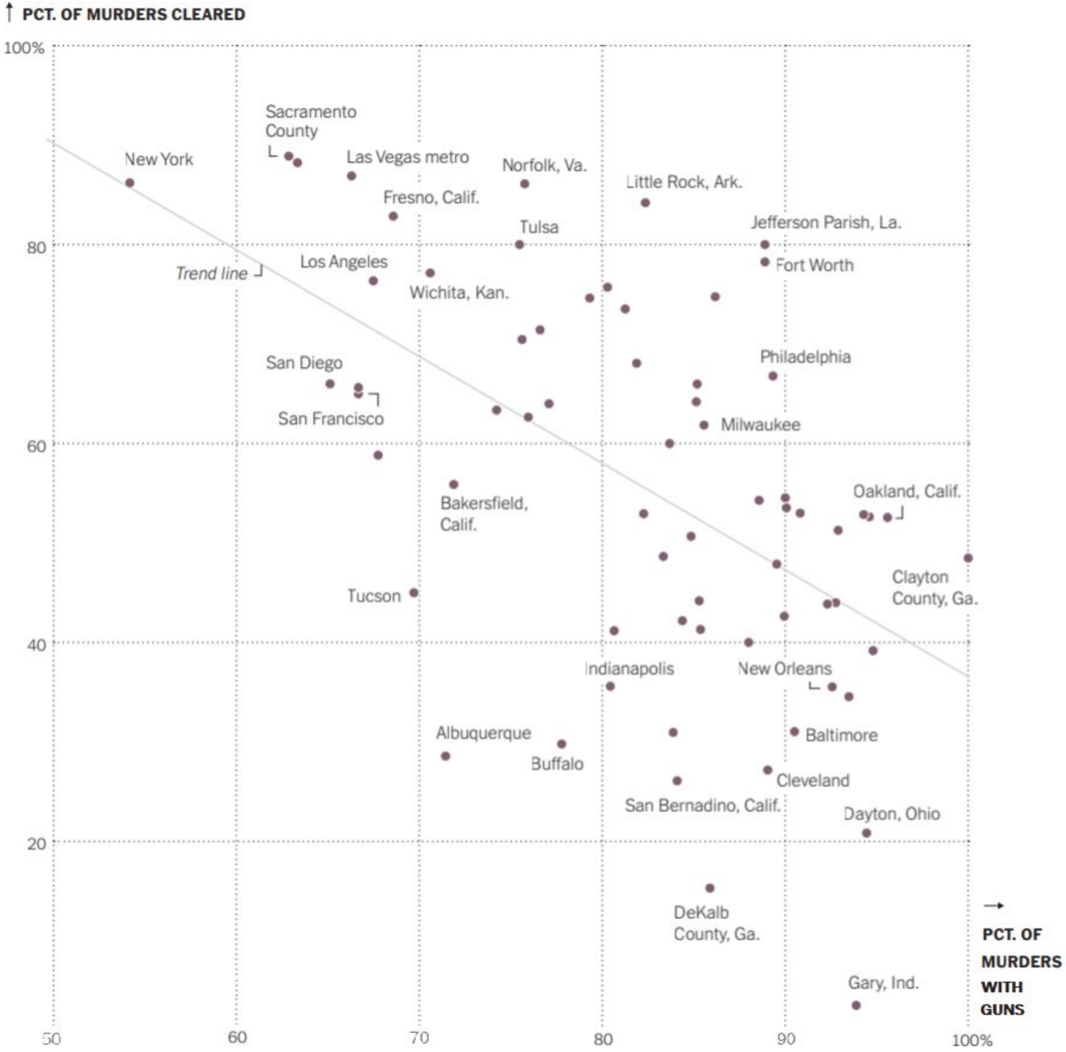
One possible confounding variable could be the rate of gang related violence in the city.

11. Link between your confounding variable and the percentage of murders committed with guns:

Gang violence is more likely to be carried out with guns

12. Link between your confounding variable and the percentage of murders that are solved:

Gang related murders are less likely to be solved than other homicides.



Group 5

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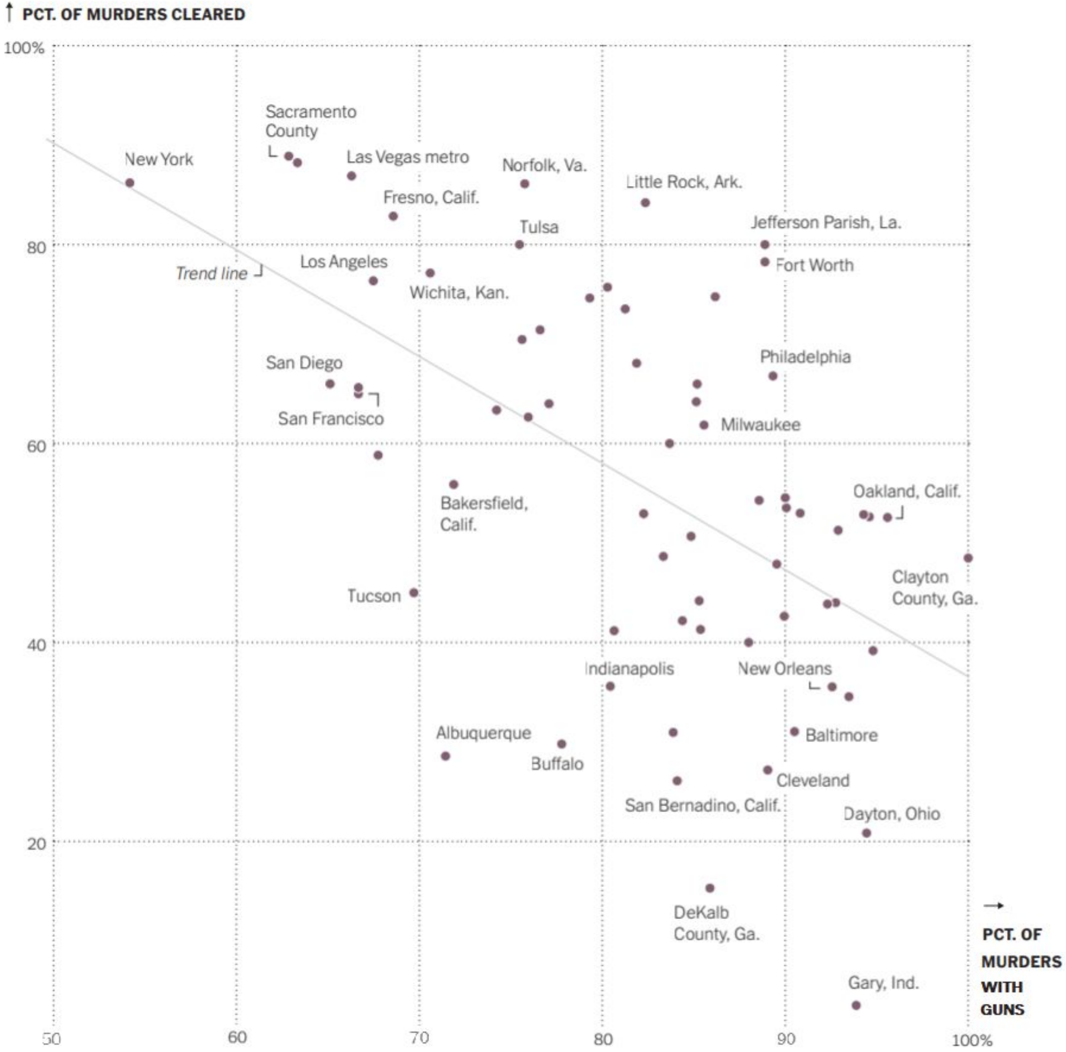


1. Tell me what you can about Norfolk, VA. Be specific.

It seems to be at the 75% of gun related murders, and at around 90% of cases solved. Meaning that guns aren't always assured, but murder cases do get solved a majority of the time.

2. Tell me what you can about New York, NY. Be specific.

New York has a gun related murder percentage of 55%, yet also seems to solve around 90% of murder cases. This means that having a gun pulled is less likely, but the same resolve rate is there compared to Norfolk.

3. Fill in the blank:

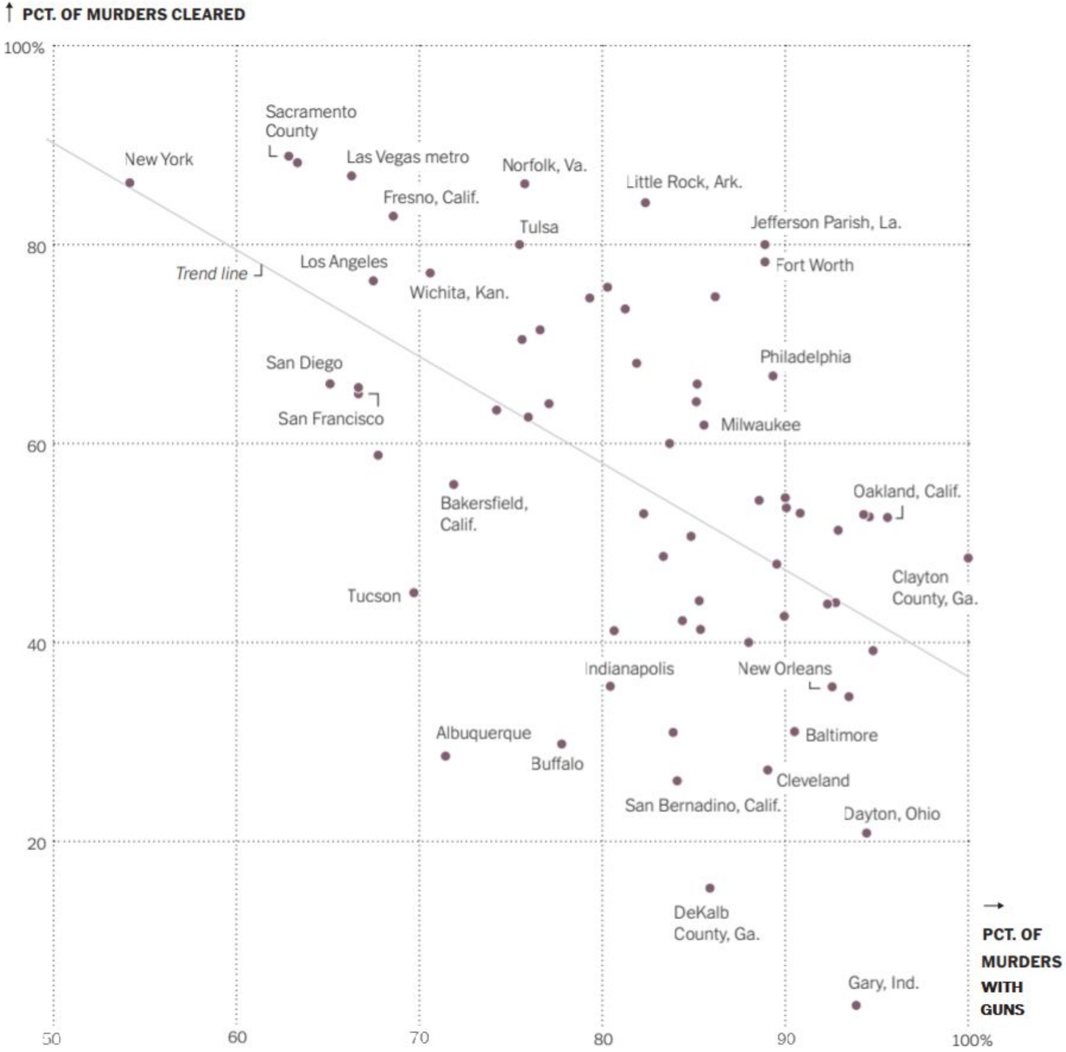
Generally speaking, for the cities represented in this graph, the greater the percentage of murders committed using a gun, the less the percentage of murder cases that are solved.

4. What is the **sample** for this visualization?

67 cities that had 30 or more murders in 2019.

5. What is the **population of interest** to the person who created this visualization, would you say?

American cities who had a reported 30 or more murders in 2019



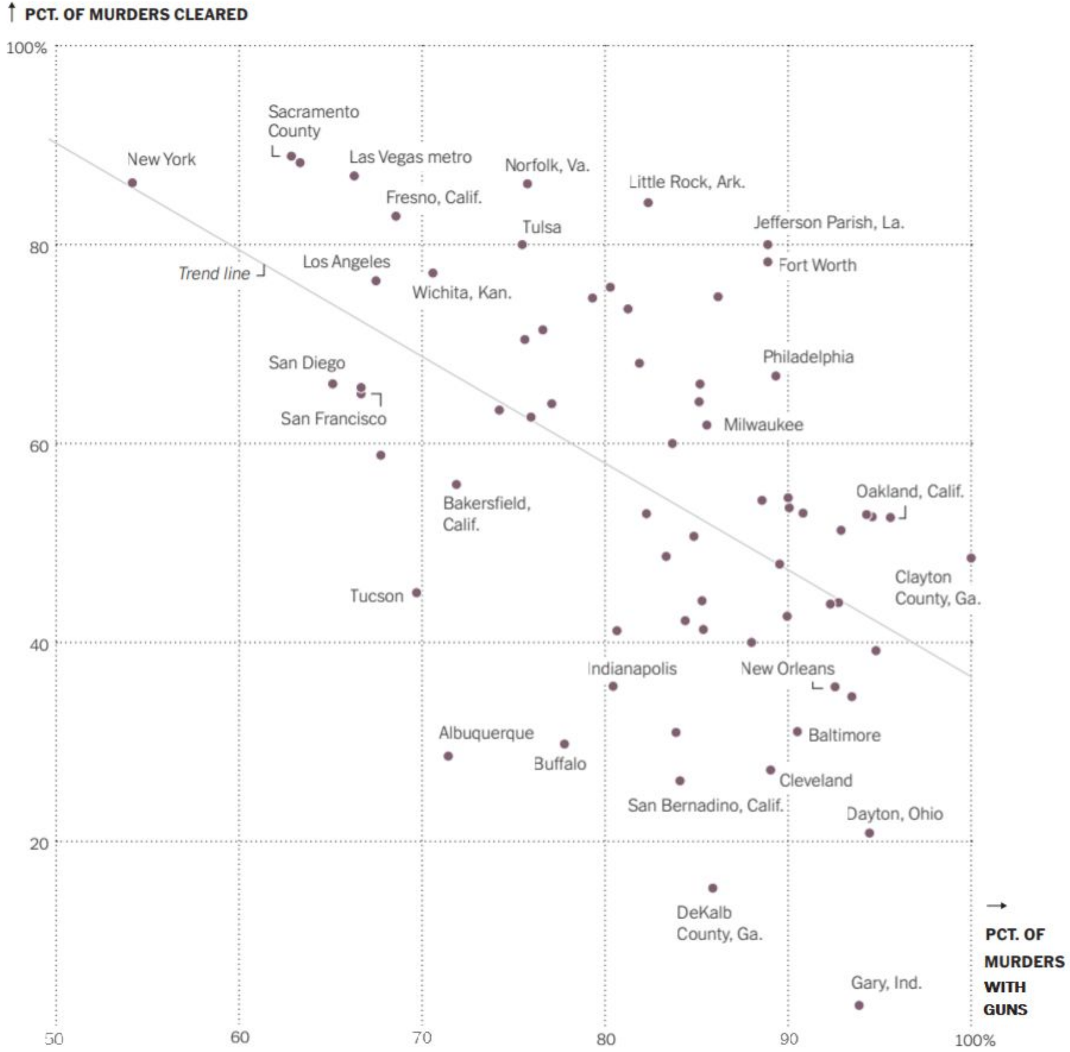
First, pretend the cases are "the murders".

6. What **variables (for cases = *murders*)** were used to create this visualization? Be careful with wording. You should have at least 3 variables.

- Was gun used?
- Was solved?
- City

7. Since each point on the graph represents a city, the cases here are actually "the cities." What **variables are recorded about each *city*** in this visualization?

Percent of murders with guns, and the percent of murders cleared.

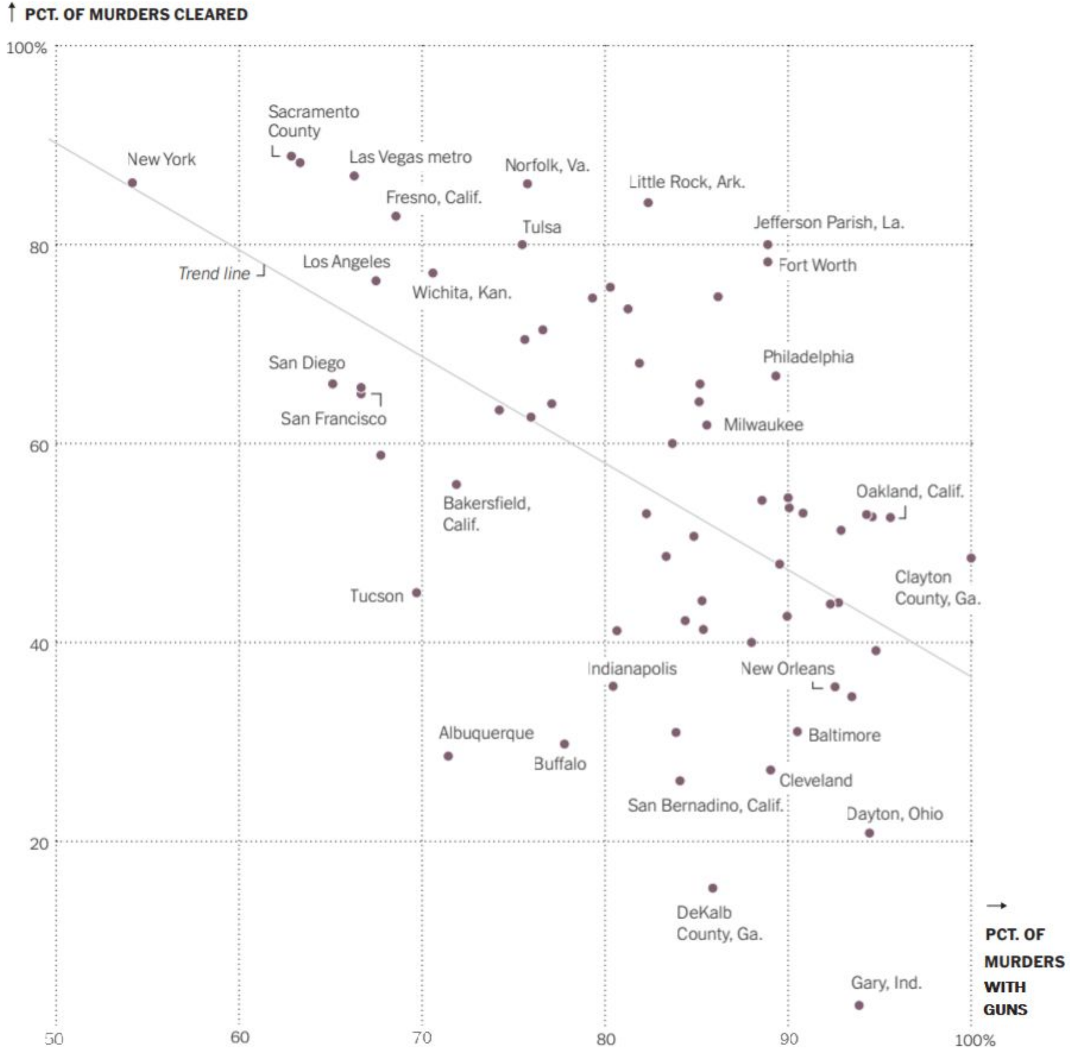


8. Can we draw the conclusion that the more murders there are committed using guns *causes* fewer murder cases to be solved? Discuss.

No, because one, this is a observational study, and thus no cause can be lifted from it. And another thing, the data points aren't closely aligned with the trendline, especially towards the 100% of murders with guns where there are far more outliers and spanning data, so you can't reasonably make the distinction.

9. Can we generalize any conclusions we draw from this graph to all cities in the U.S.? Discuss.

I don't think so because the cut off for cities included was high, meaning that smaller cities or cities with a lower crime rate will not be included. Additionally, the cities were not randomly sampled.



10. Propose a possible confounding variable that might explain the association between percentage of murders with guns and percentage of murders cleared.

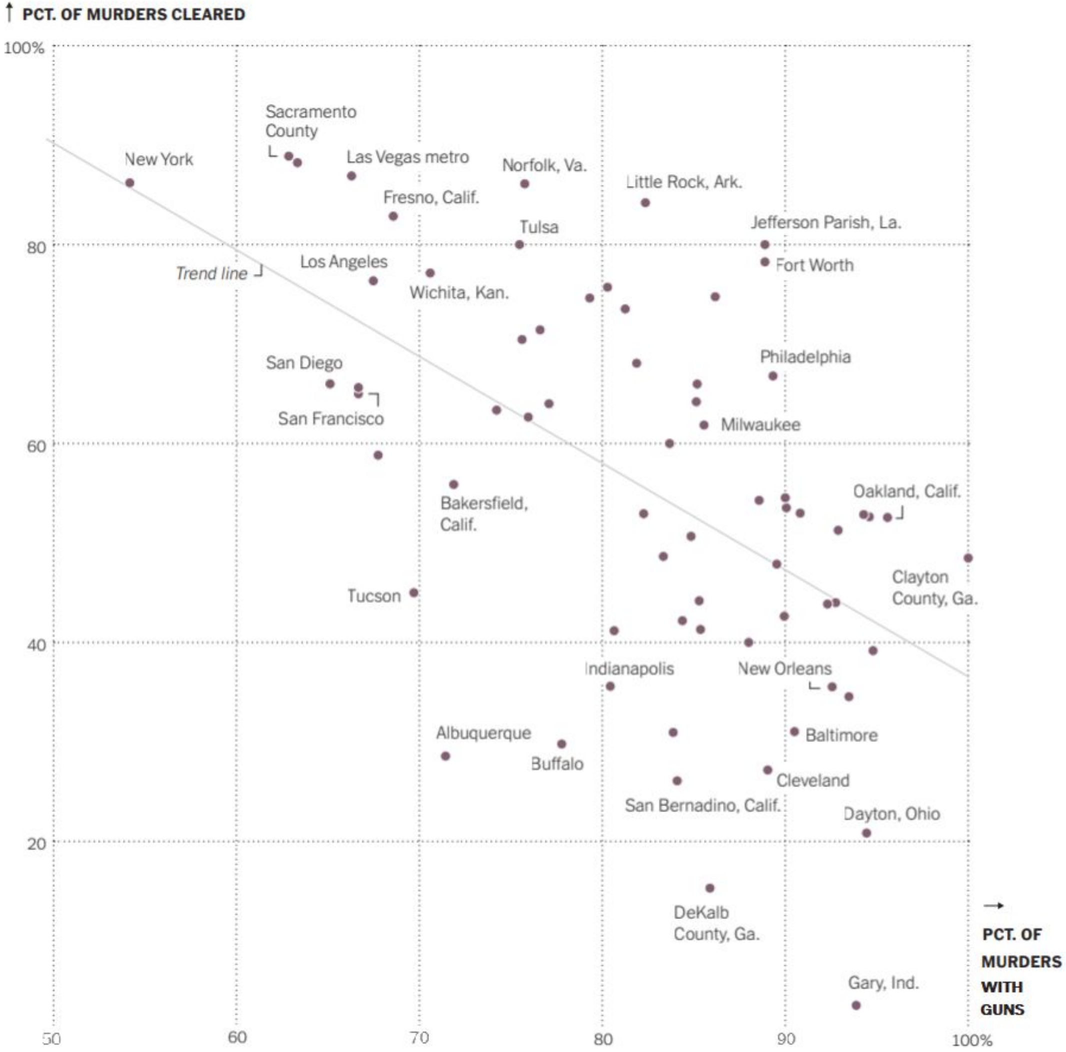
A non gun weapon tends to leave more easily traceable evidence behind, meanwhile for guns, it's harder to do things like say who was on the scene, because some guns might not need to be on the scene at all, and thus a lot of identifying evidence might be lost because of that distance.

11. Link between your confounding variable and the percentage of murders committed with guns:

Guns are a more readily reliable weapon type, especially for planned murders, and so is more likely to be used.

12. Link between your confounding variable and the percentage of murders that are solved:

Guns tend to leave less identifying information behind like fingerprints, and sometimes the person might not even need to be on the scene, so it's much easier to get rid of the weapon, thus losing much evidence. You know when you've been stabbed by a blade. You don't often know what kind of gun someone is holding.



Group 6

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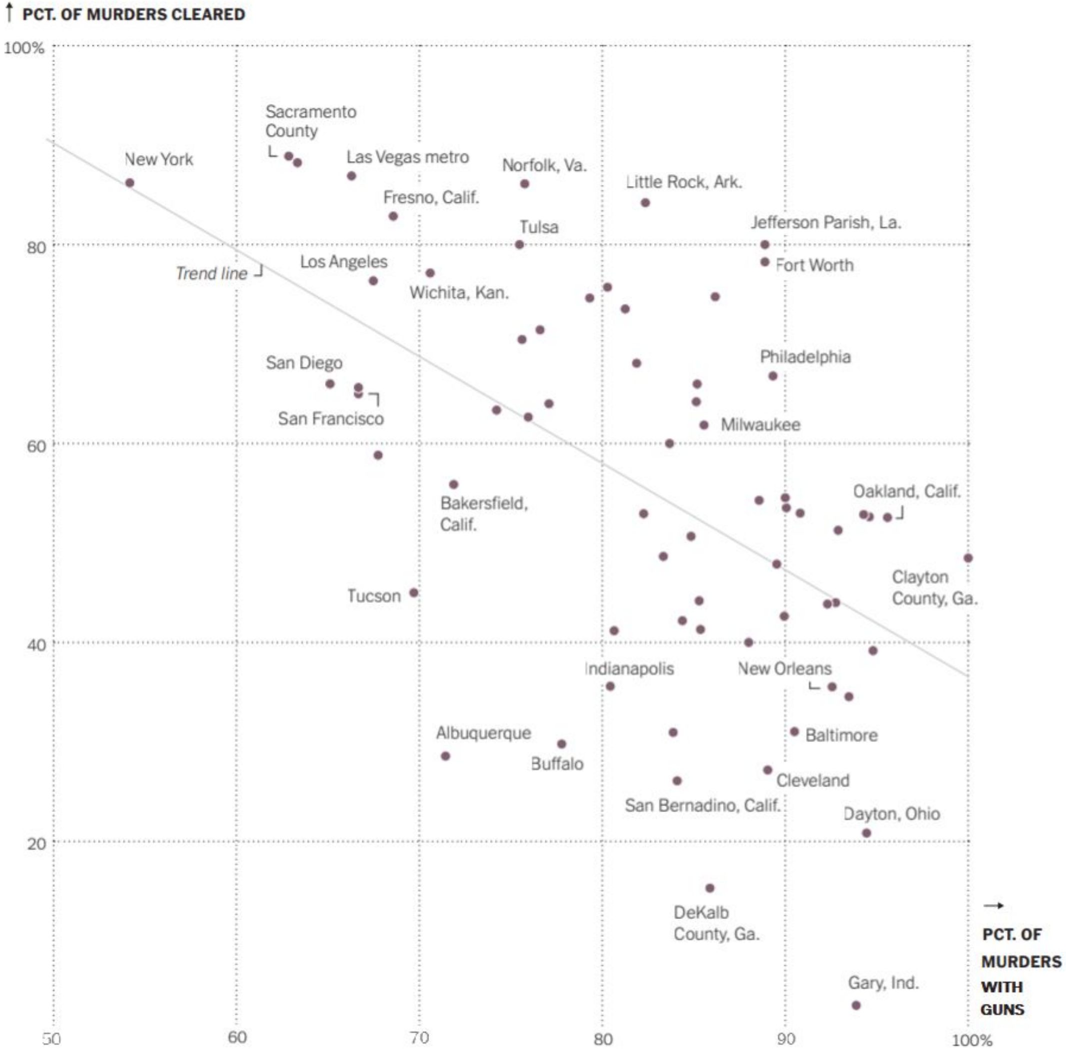
This graph shows the relationship between the percentage of murders committed with guns for the 67 cities that had 30 or more murders in 2019 and the percentage of all murders solved."



1. Tell me what you can about Norfolk, VA. Be specific.

Approximately 76% of the murders in Norfolk are committed with guns, and approximately 85% of the murders in Norfolk are cleared.

2. Tell me what you can about New York, NY. Be specific.

NYNY has about 55% of its murders committed with guns, but has a clearance rate of around 85%

3. Fill in the blank:

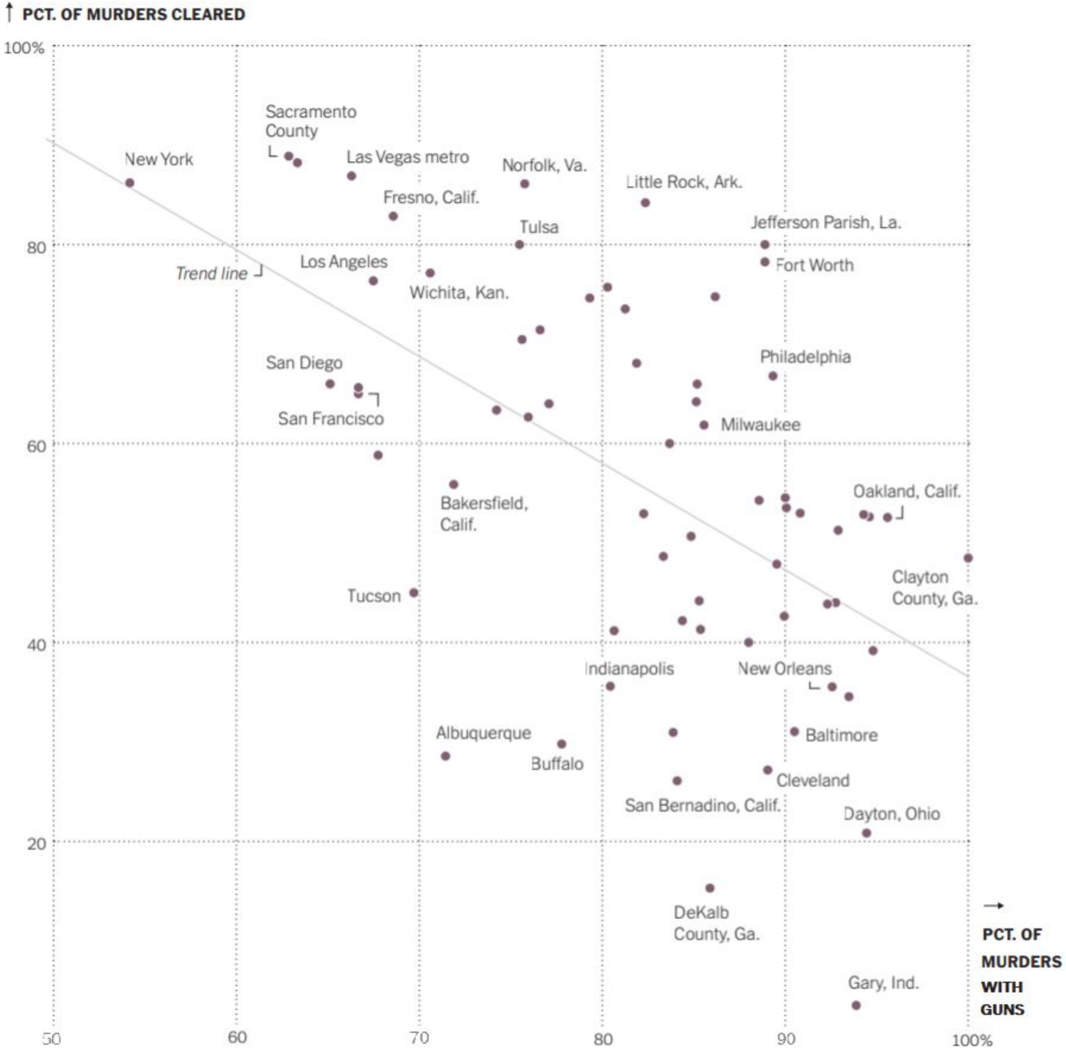
Generally speaking, for the cities represented in this graph, the greater the percentage of murders committed using a gun, the <u>smaller </u>the percentage of murder cases that are solved.

4. What is the **sample** for this visualization?

Reported/documented murder rates in 67 cities with 30 or more murders

5. What is the **population of interest** to the person who created this visualization, would you say?

Miurder cases in America



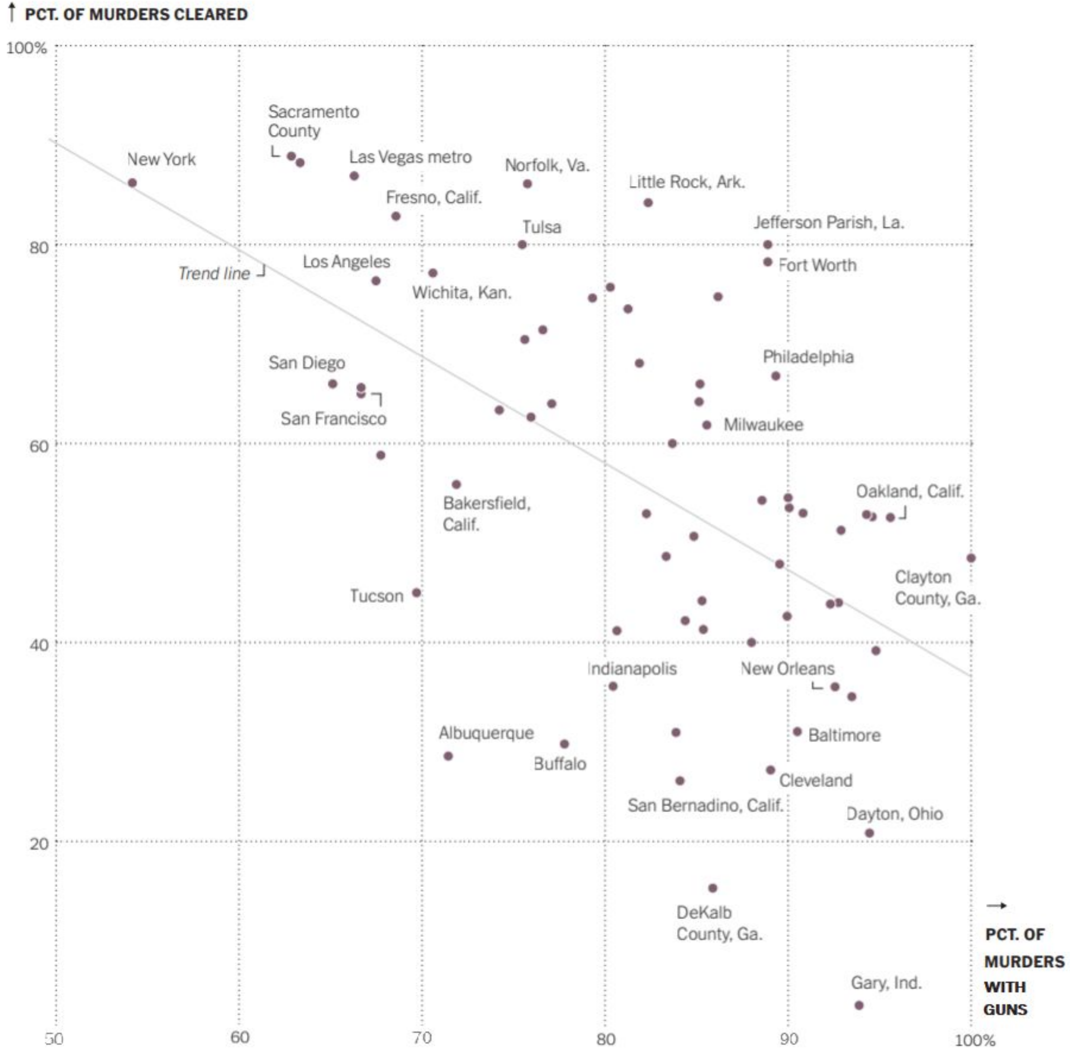
First, pretend the cases are “the murders”.

6. What **variables (for cases = *murders*)** were used to create this visualization? Be careful with wording. You should have at least 3 variables.

- Likelihood of being cleared
- City the murders took place in
- murdered with a gun

7. Since each point on the graph represents a city, the cases here are actually “the cities.” What **variables are recorded about each *city*** in this visualization?

- % of murders cleared
- % of murders committed with guns

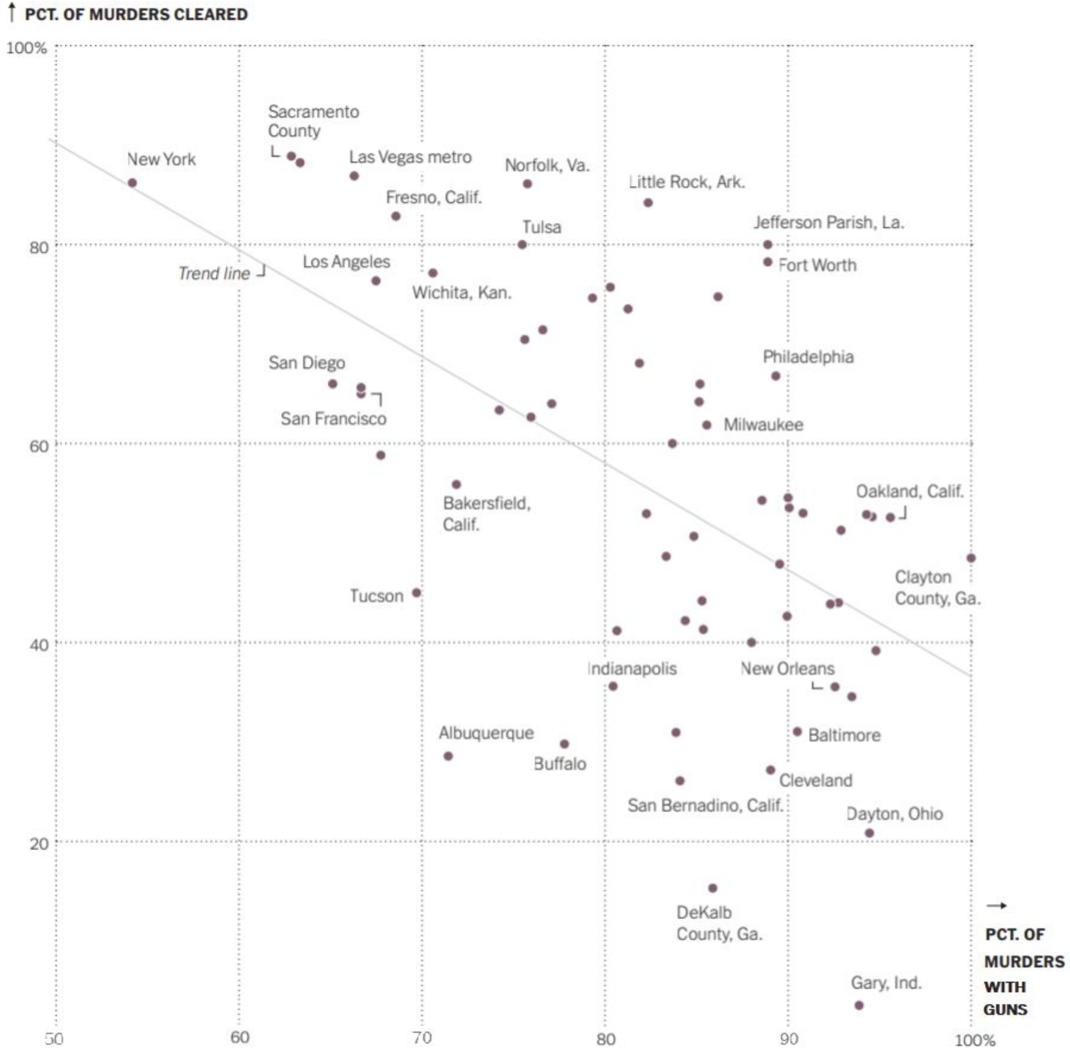


8. Can we draw the conclusion that the more murders there are committed using guns *causes* fewer murder cases to be solved? Discuss.

No, this is an observational study. Causality is not able to be determined from this.

9. Can we generalize any conclusions we draw from this graph to all cities in the U.S.? Discuss.

Generally, the rate in which murders are cleared decreases in a city as the proportion of murders with guns increases



10. Propose a possible confounding variable that might explain the association between percentage of murders with guns and percentage of murders cleared.

Geographic region of the city
Rates of other violent crimes in a given city

11. Link between your confounding variable and the percentage of murders committed with guns:

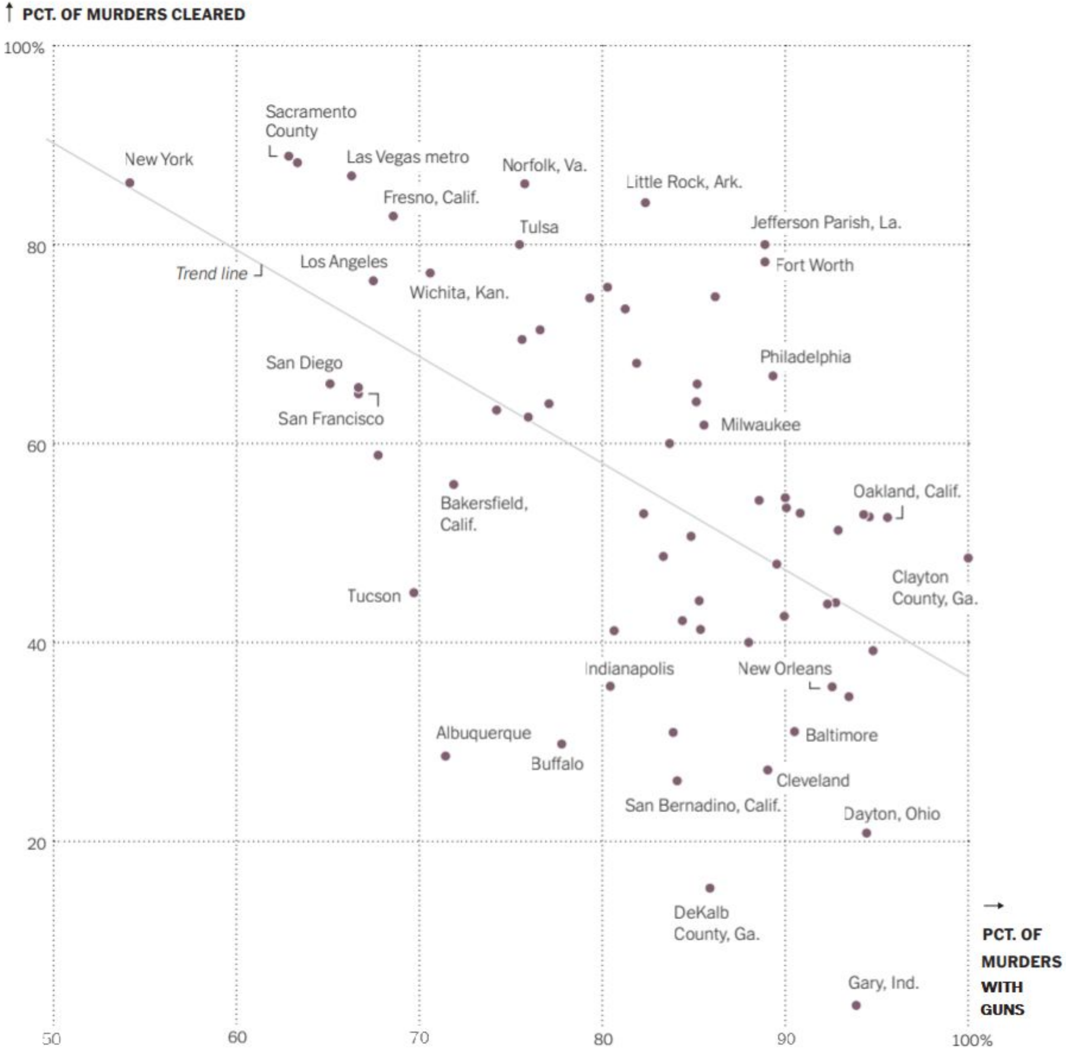
Certain regions have higher gun violence rates than others

A city with a higher violent crime may have a higher gun violence rate

12. Link between your confounding variable and the percentage of murders that are solved:

More violent crimes in general means that there are less crimes able to be cleared due to sheer volume/concentration of police effort.

Certain regions have more effective police forces than others



Group 7

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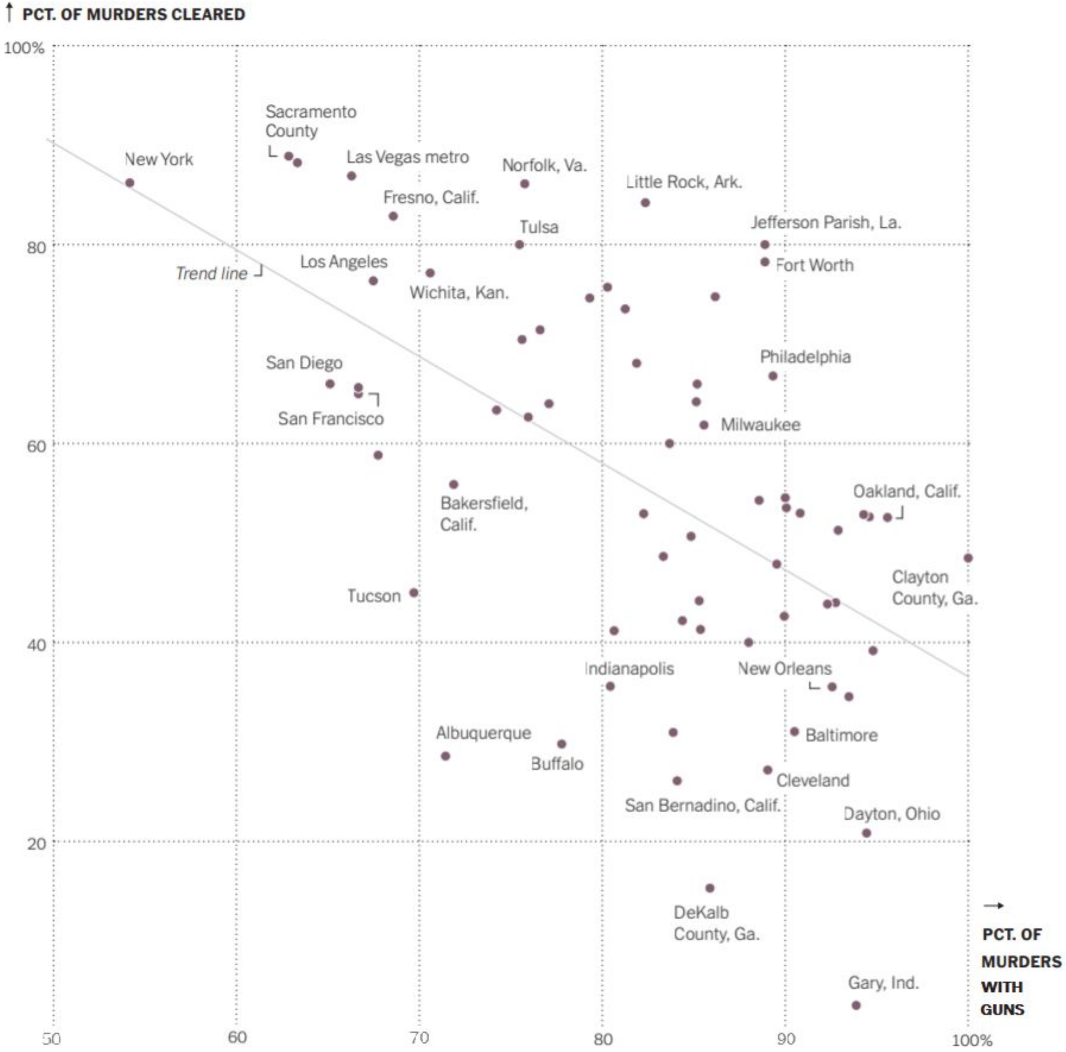


1. Tell me what you can about Norfolk, VA. Be specific.

With Norfolk Virginia about 76% of murders are committed with guns. But the murders cleared are 86%.

2. Tell me what you can about New York, NY. Be specific.

New York has the least amount of murders committed with a gun at about 54%. Also New York has one of the highest cleared percentages.

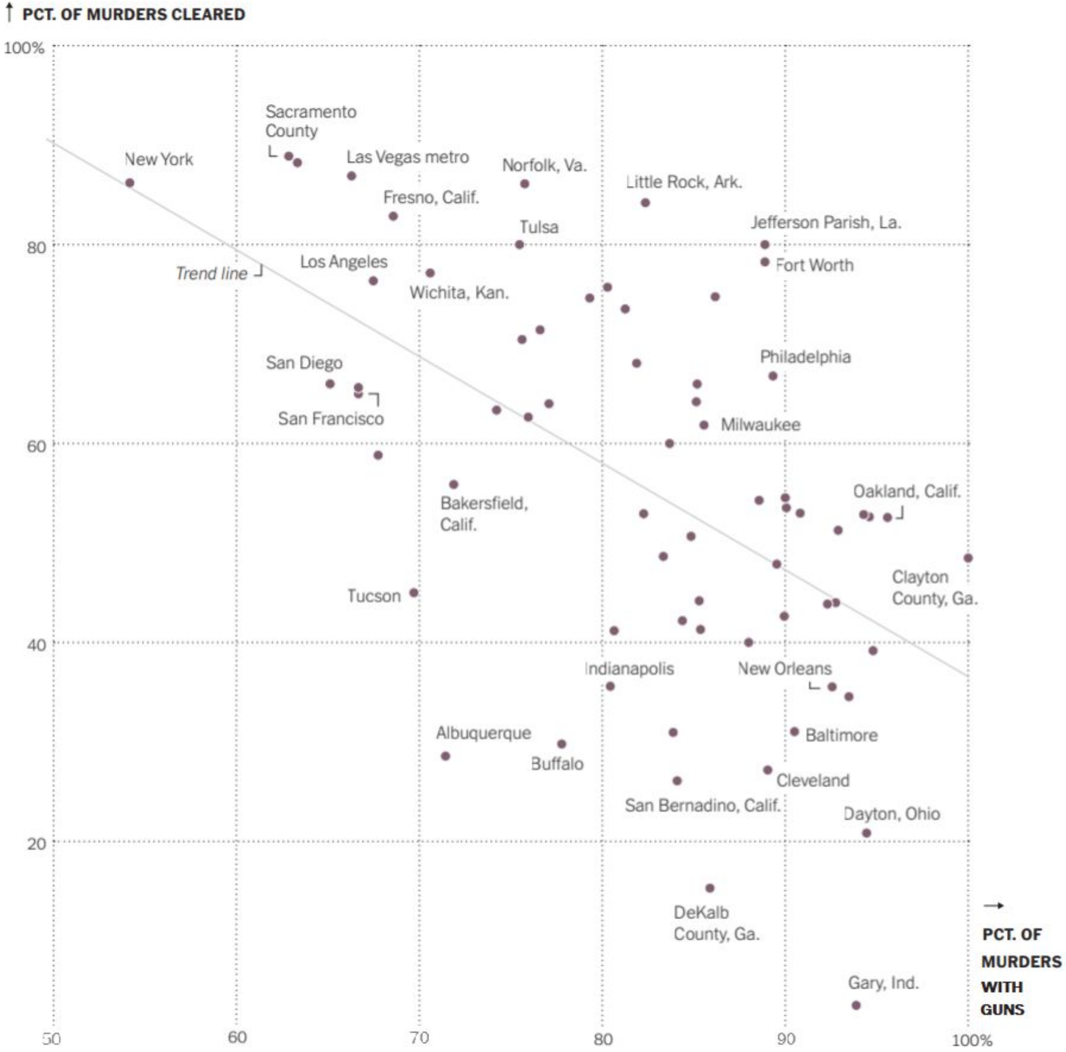
3. Fill in the blank:

Generally speaking, for the cities represented in this graph, the greater the percentage of murders committed using a gun, the _higher__ the percentage of murder cases that are solved.

4. What is the **sample** for this visualization?

Number of cities (assuming the cases are all in the US)

5. What is the **population of interest** to the person who created this visualization, would you say? Percentage of cleared



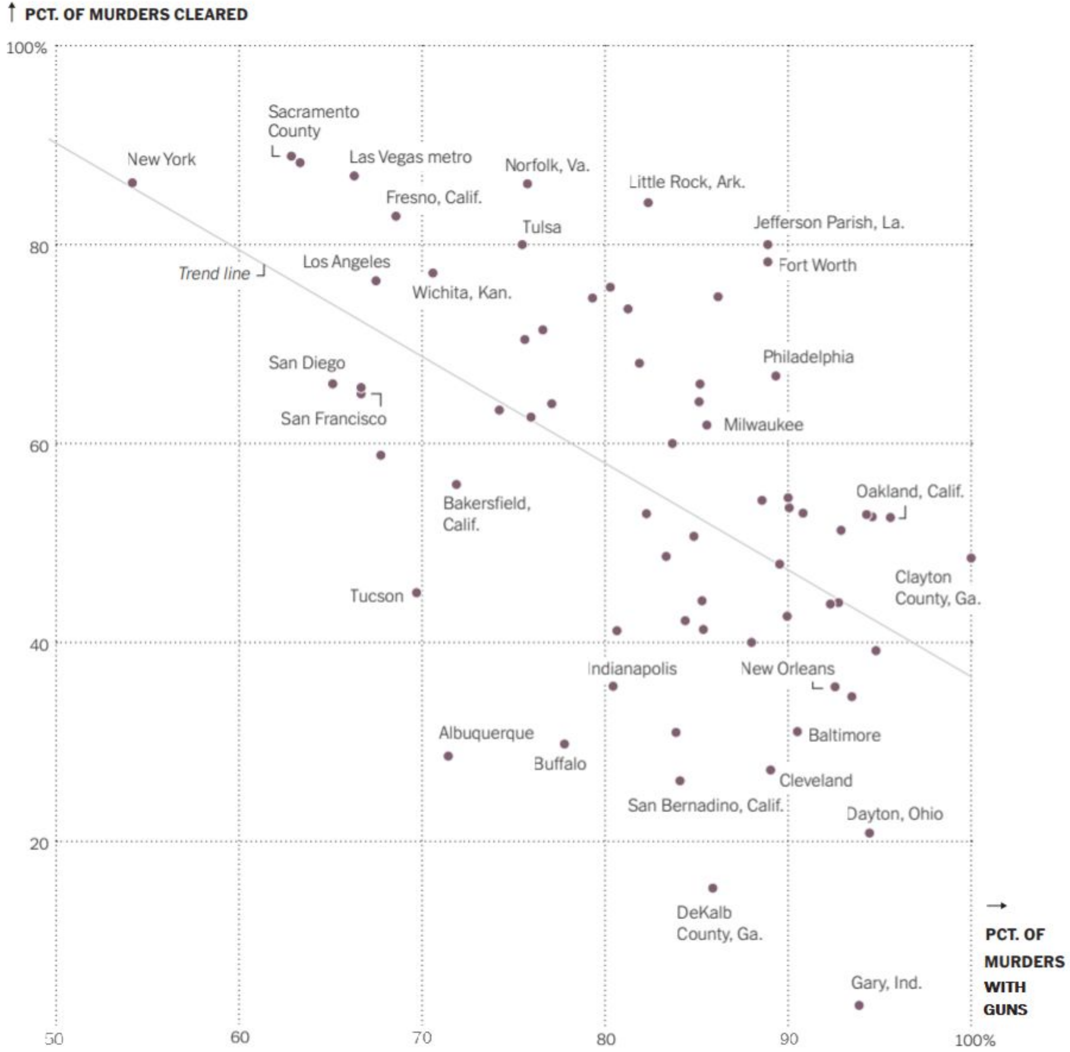
First, pretend the cases are “the murders”.

6. What **variables (for cases = *murders*)** were used to create this visualization? Be careful with wording. You should have at least 3 variables.

Murders with guns, other weapons used, amount of murders cleared

7. Since each point on the graph represents a city, the cases here are actually “the cities.” What **variables are recorded about each *city*** in this visualization?

Cases cleared and murders with guns

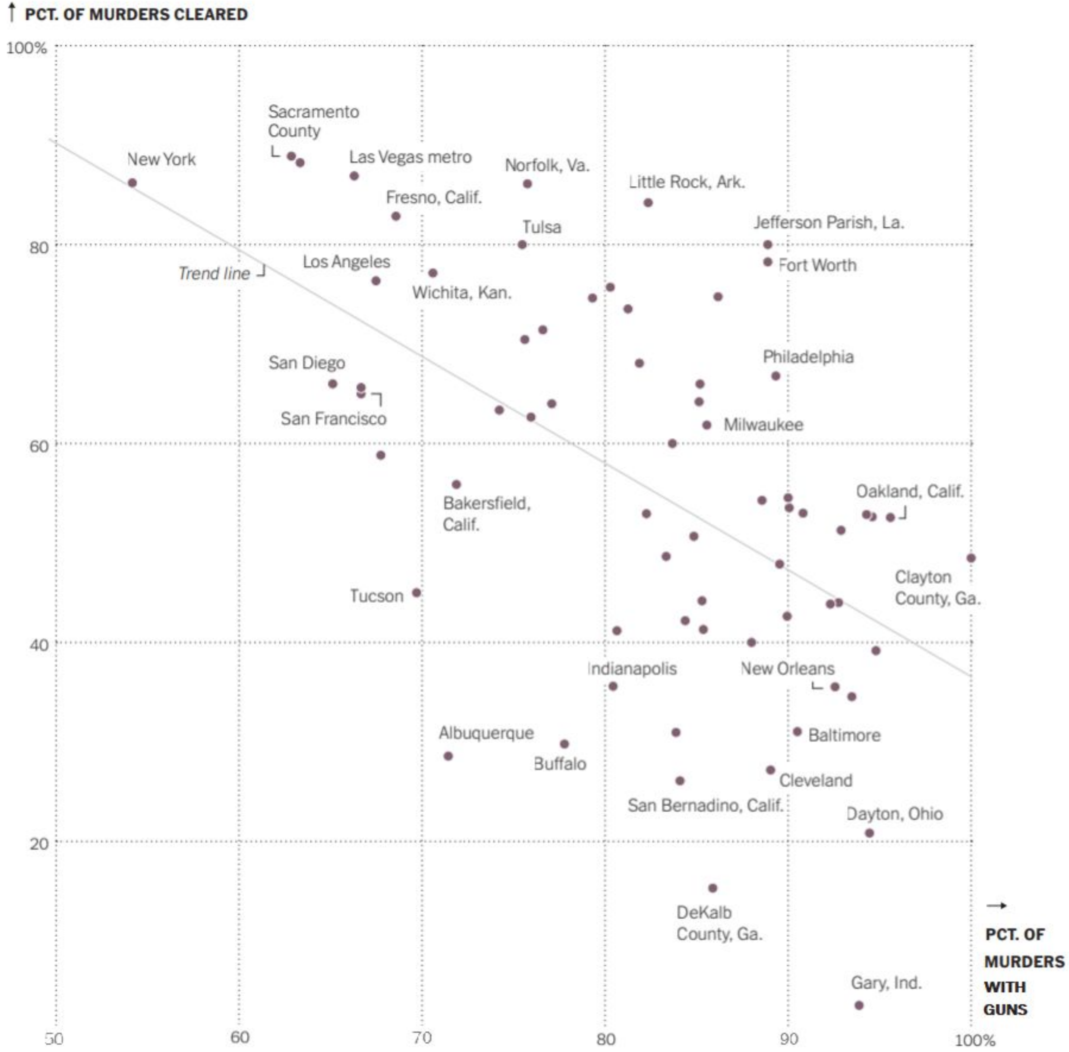


8. Can we draw the conclusion that the more murders there are committed using guns *causes* fewer murder cases to be solved? Discuss.

We can associate the variable of murders with guns and the amount of cases cleared, but we can not say it causes fewer cases solved

9. Can we generalize any conclusions we draw from this graph to all cities in the U.S.? Discuss.

In this case we can not generalize the conclusions with other cities that may not be recorded because the cities are not randomly selected.



10. Propose a possible confounding variable that might explain the association between percentage of murders with guns and percentage of murders cleared.

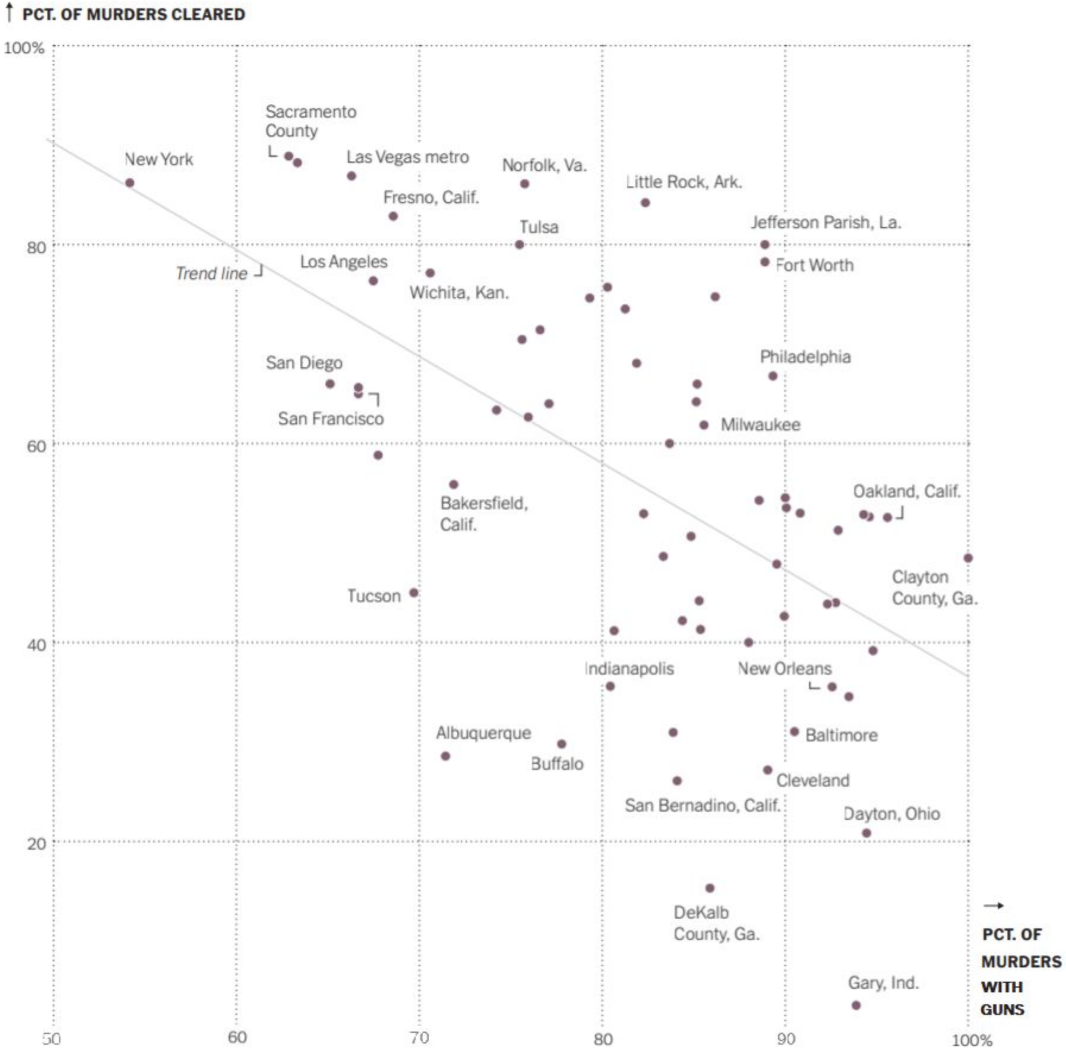
Level of experience an individual has

11. Link between your confounding variable and the percentage of murders committed with guns:

The link between the confounding variable of personnel that are on the cases is the experience that the people have that are on the job, someone with more experience or have seen a certain situation before could solve a murder easier or solve it in general better than someone new.

12. Link between your confounding variable and the percentage of murders that are solved:

The link between level of experience and the percentage of murders solved is some cities may have more experienced police officers and personnel that are able to clear cases faster than other locations.

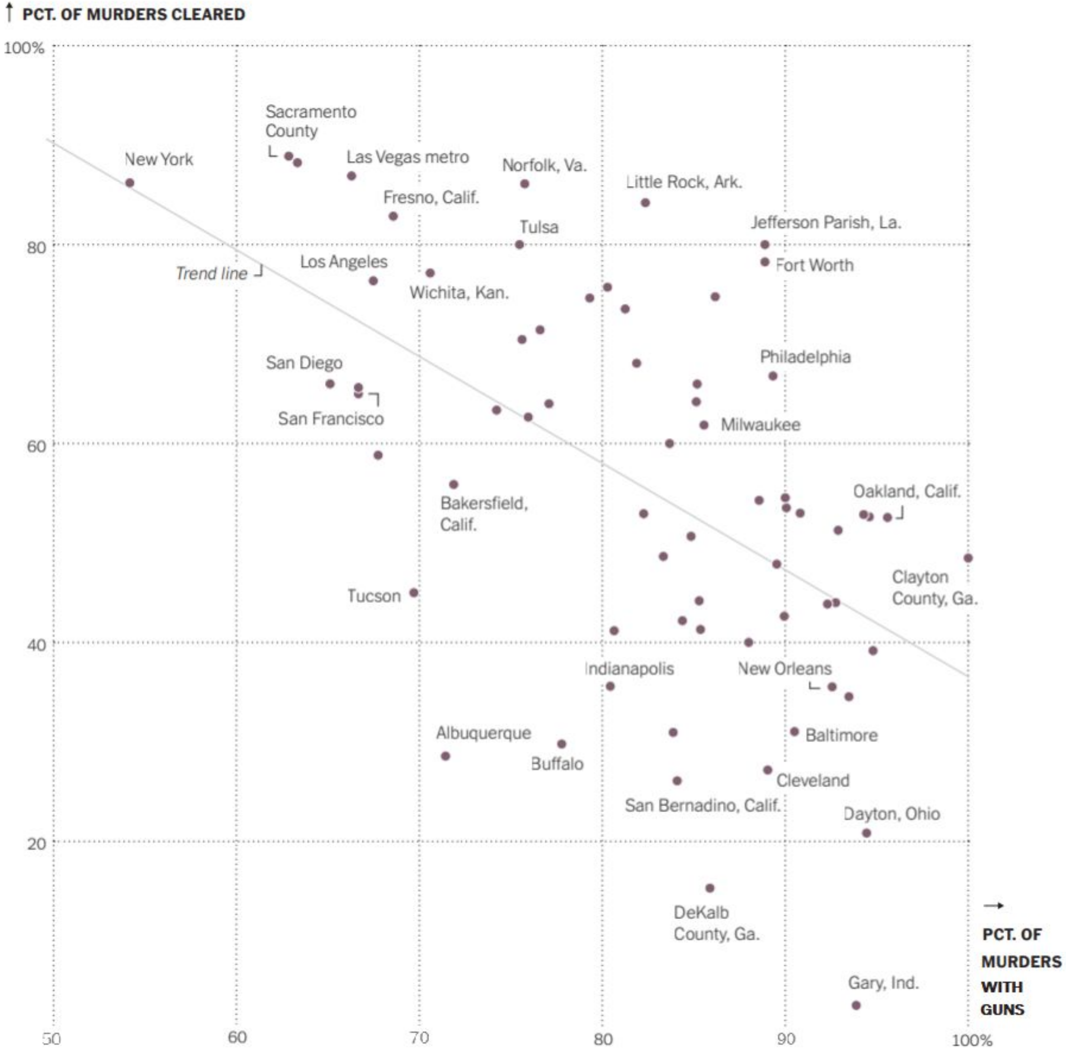


Group 8

This graph is from the NYTimes. Please don't search for the article just yet. Here is what the opening of the article says about the graph:

"Is the likelihood of solving a murder case less if the weapon is a gun?

This graph shows the relationship between the percentage of murders committed with guns for the 67 cities that had 30 or more murders in 2019 and the percentage of all murders solved."



1. Tell me what you can about Norfolk, VA. Be specific.

Norfolk has 86% of its murders with guns, and 88% of its murders cleared.

2. Tell me what you can about New York, NY. Be specific.

New York has 54% of its murders with guns, and 88% of its murders cleared.

3. Fill in the blank:

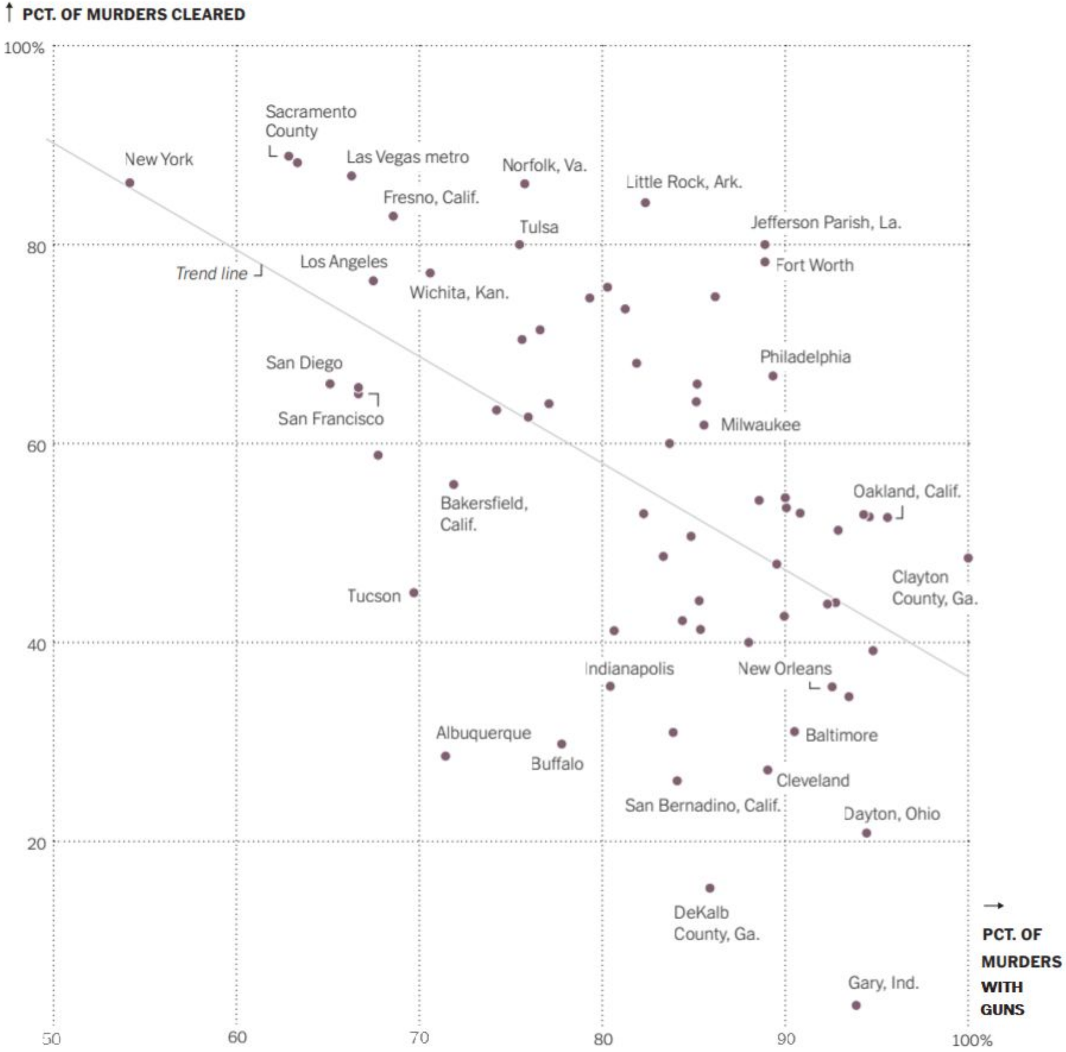
Generally speaking, for the cities represented in this graph, the greater the percentage of murders committed using a gun, the **lower** the percentage of murder cases that are solved.

4. What is the **sample** for this visualization?

The sample for this visualization is 67 cities that had 30 or more murders in 2019.

5. What is the **population of interest** to the person who created this visualization, would you say?

The population of interest are US cities with a high number of murders.



First, pretend the cases are "the murders".

6. What **variables (for cases = *murders*)** were used to create this visualization? Be careful with wording. You should have at least 3 variables.

Murder cleared? (y/n)

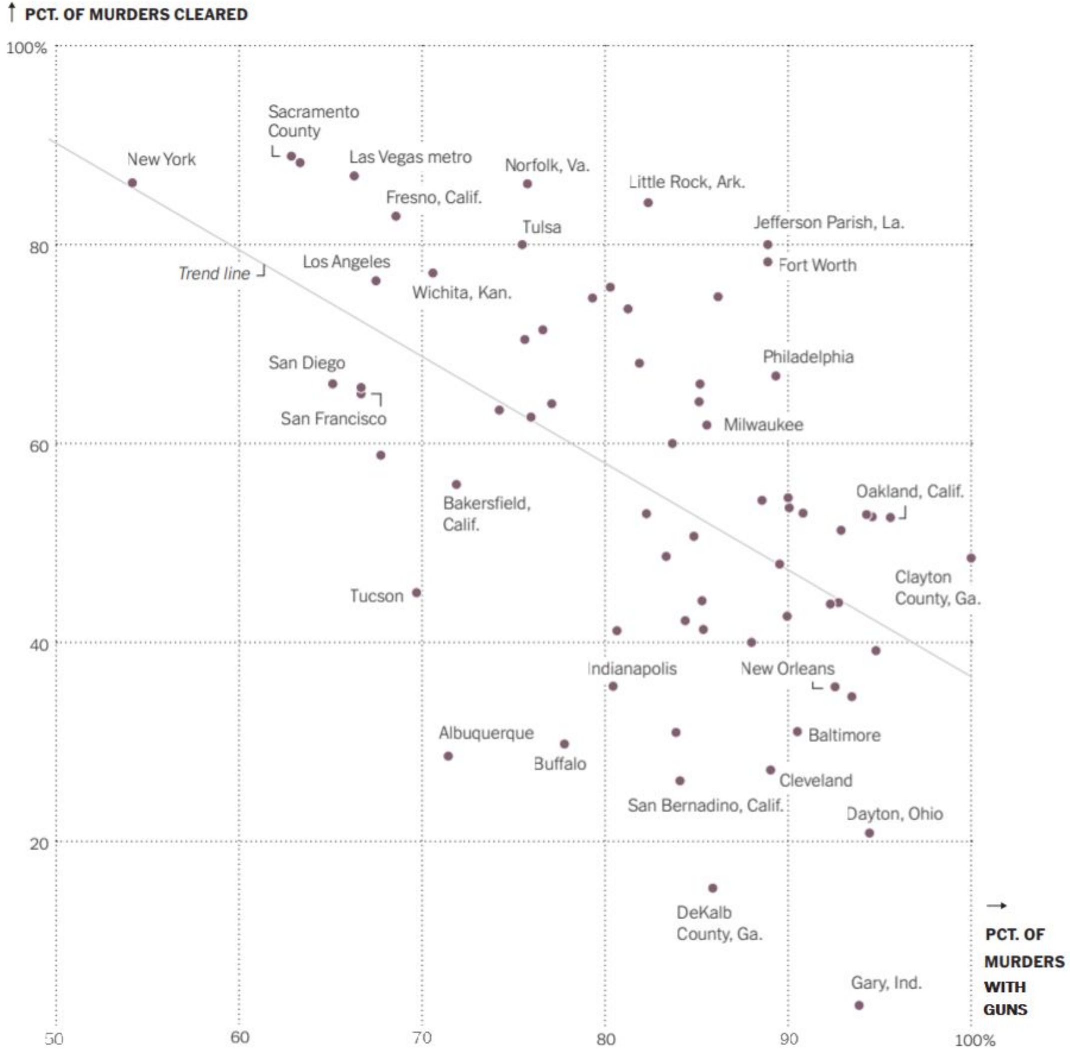
Murder with a gun? (y/n)

City of murder

7. Since each point on the graph represents a city, the cases here are actually "the cities." What **variables are recorded about each *city*** in this visualization?

% of murders cleared

% of murders with guns

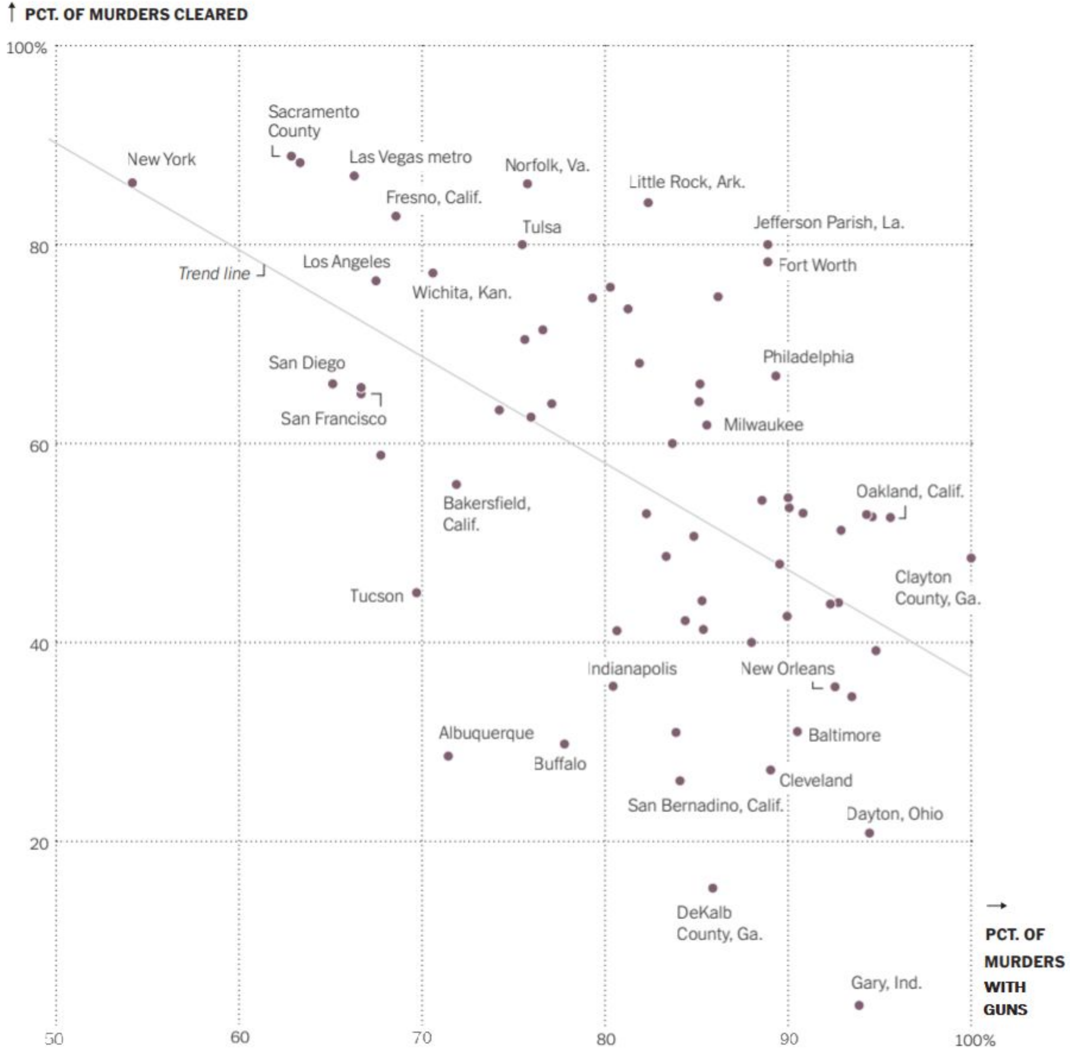


8. Can we draw the conclusion that the more murders there are committed using guns *causes* fewer murder cases to be solved? Discuss.

We can conclude association, but not causation because this is an observational study.

9. Can we generalize any conclusions we draw from this graph to all cities in the U.S.? Discuss.

We cannot generalize because it isn't a random sample of all U.S. cities; cities with less than 30 murders are not included.



10. Propose a possible confounding variable that might explain the association between percentage of murders with guns and percentage of murders cleared.

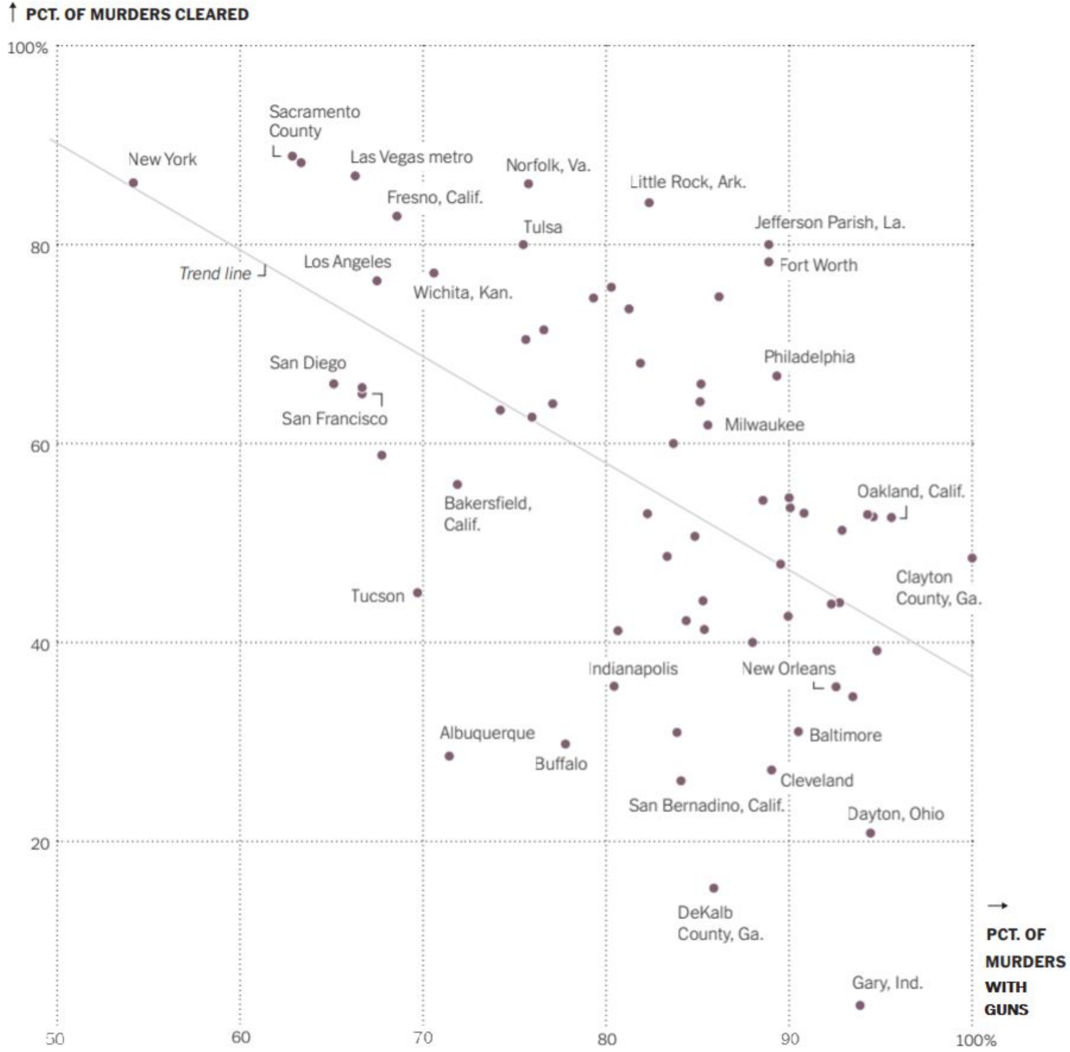
A possible confounding variable would be whether the state is red or blue.

11. Link between your confounding variable and the percentage of murders committed with guns:

Red cities often have less gun control, and more guns per capita which could lead to more gun related murders. Access to guns would be easier in red cities, so more murders likely involve guns.

12. Link between your confounding variable and the percentage of murders that are solved:

Red cities typically employ fewer police than blue cities, which would mean blue cities might have an easier time and more resources to solve murders, while red cities may have a harder time with staffing and resources to solve murders. Red states also have an average of 168 cops per 100k, while blue cities have 260 cops per 100k. Blue states also spend much more on policing than red states. This could lead to more murders being unsolved due to understaffing and lack of resources.

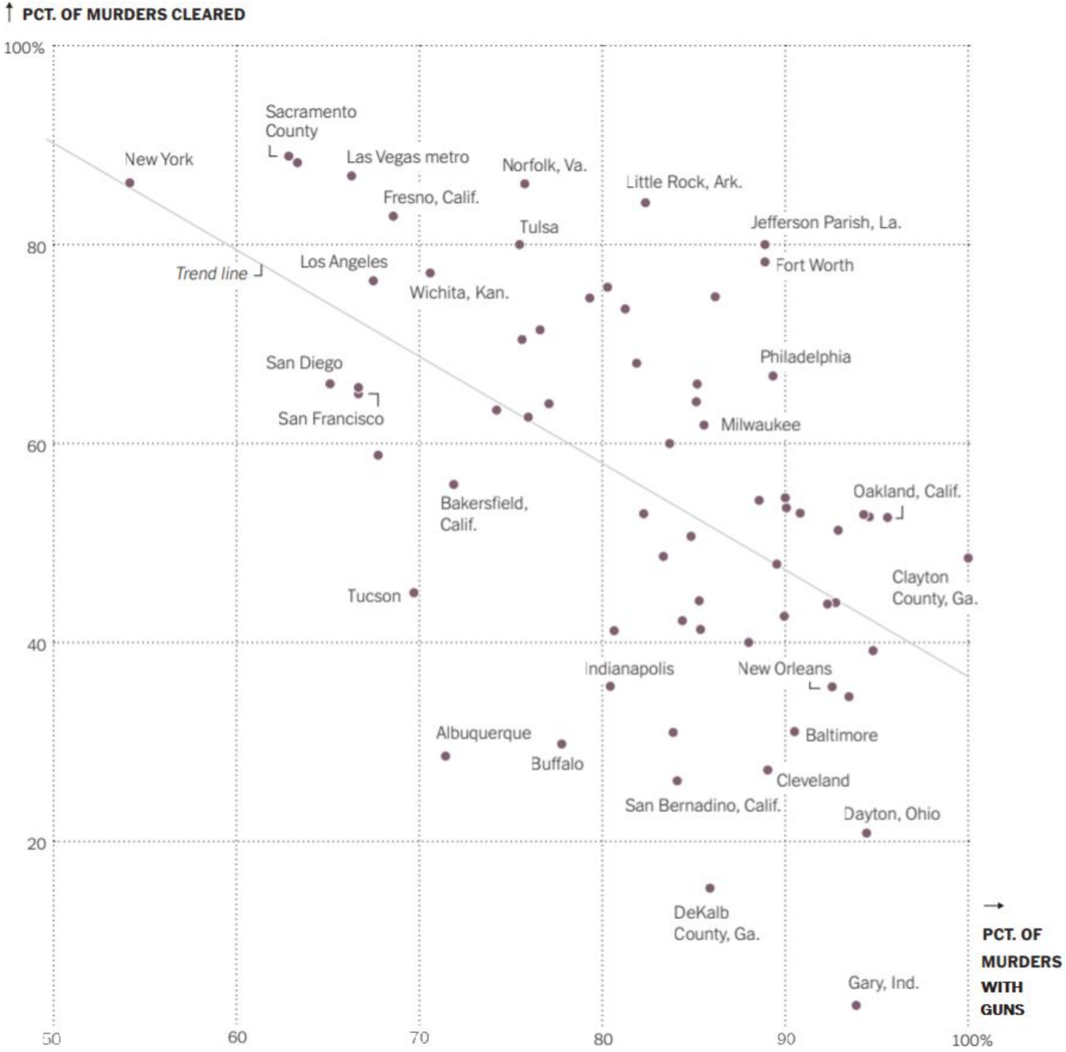


Group 9

This graph is from the NYTimes. Please don't search for the article just yet. Here is what the opening of the article says about the graph:

"Is the likelihood of solving a murder case less if the weapon is a gun?

This graph shows the relationship between the percentage of murders committed with guns for the 67 cities that had 30 or more murders in 2019 and the percentage of all murders solved."



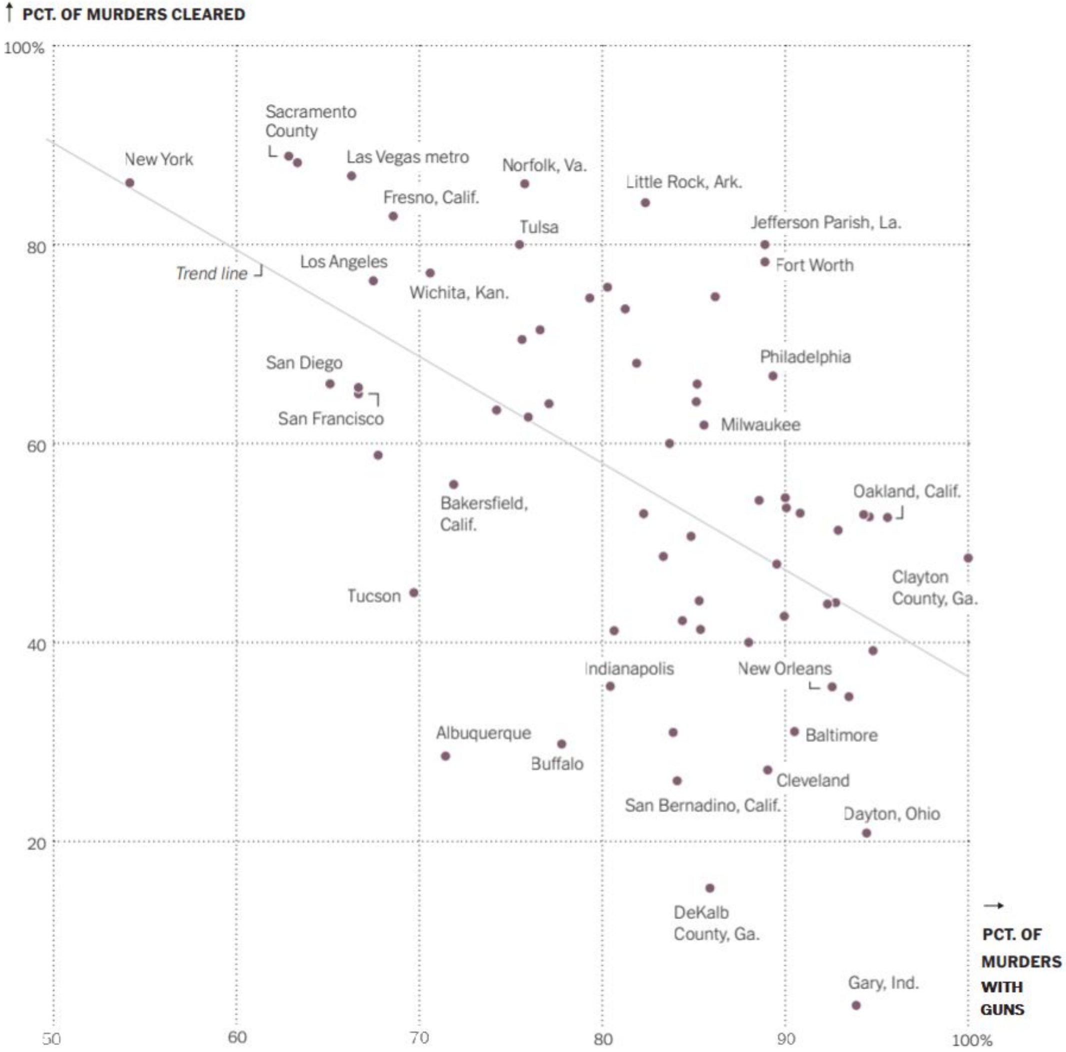
1. Tell me what you can about Norfolk, VA. Be specific.
    a. Norfolk has a high percentage of murders cleared with a 75% percentage of their murders committed with a firearm.

2. Tell me what you can about New York, NY. Be specific.
    a. New York has about 55% of murders committed with a gun but also has a high percentage of murders cleared.
3. Fill in the blank:

Generally speaking, for the cities represented in this graph, the greater the percentage of murders committed using a gun, the **lower** the percentage of murder cases that are solved.

4. What is the **sample** for this visualization?
    a. 67 cities that had 30 or more murders in 2019

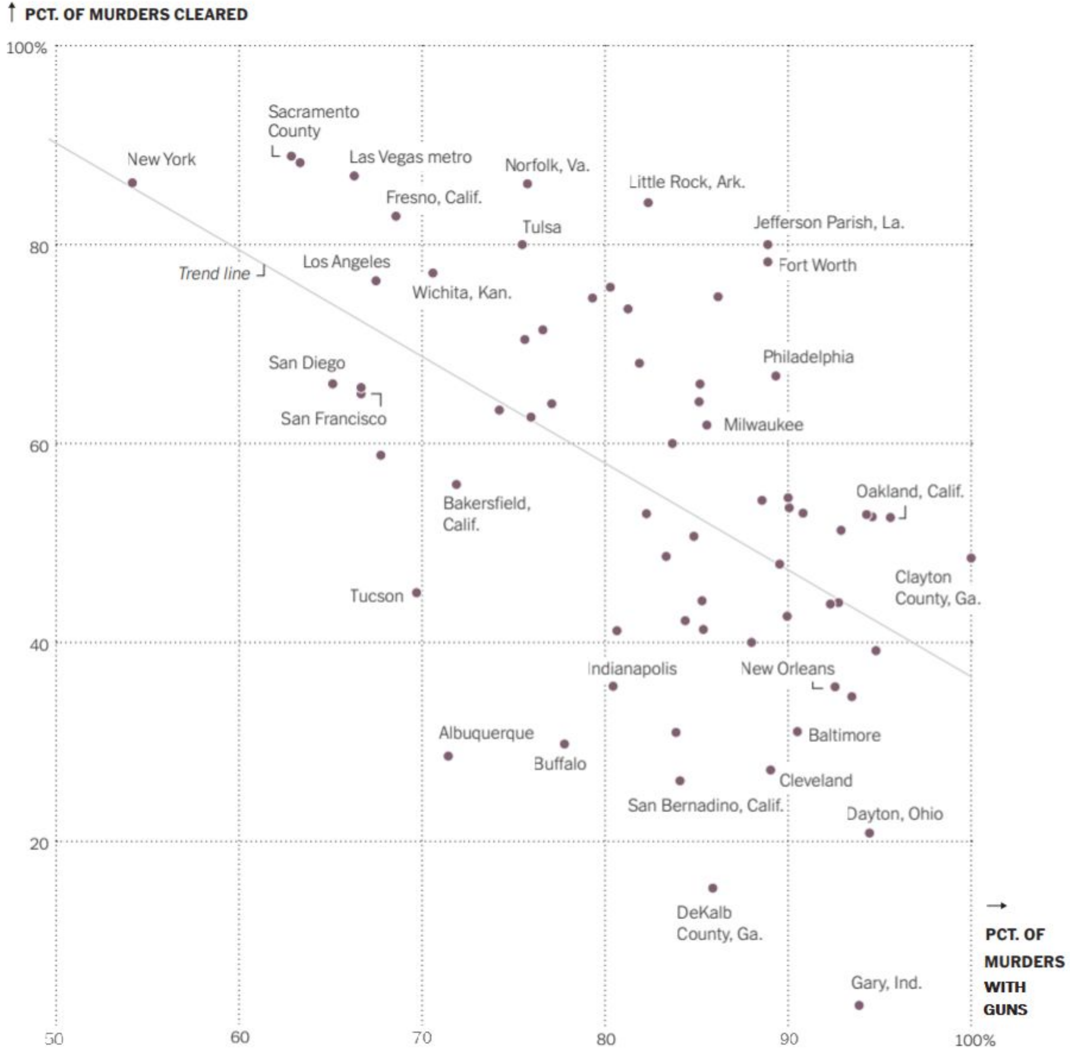
5. What is the **population of interest** to the person who created this visualization, would you say?
    a. US Cities with more than 30 murders a year (for 2019)



First, pretend the cases are “the murders”.

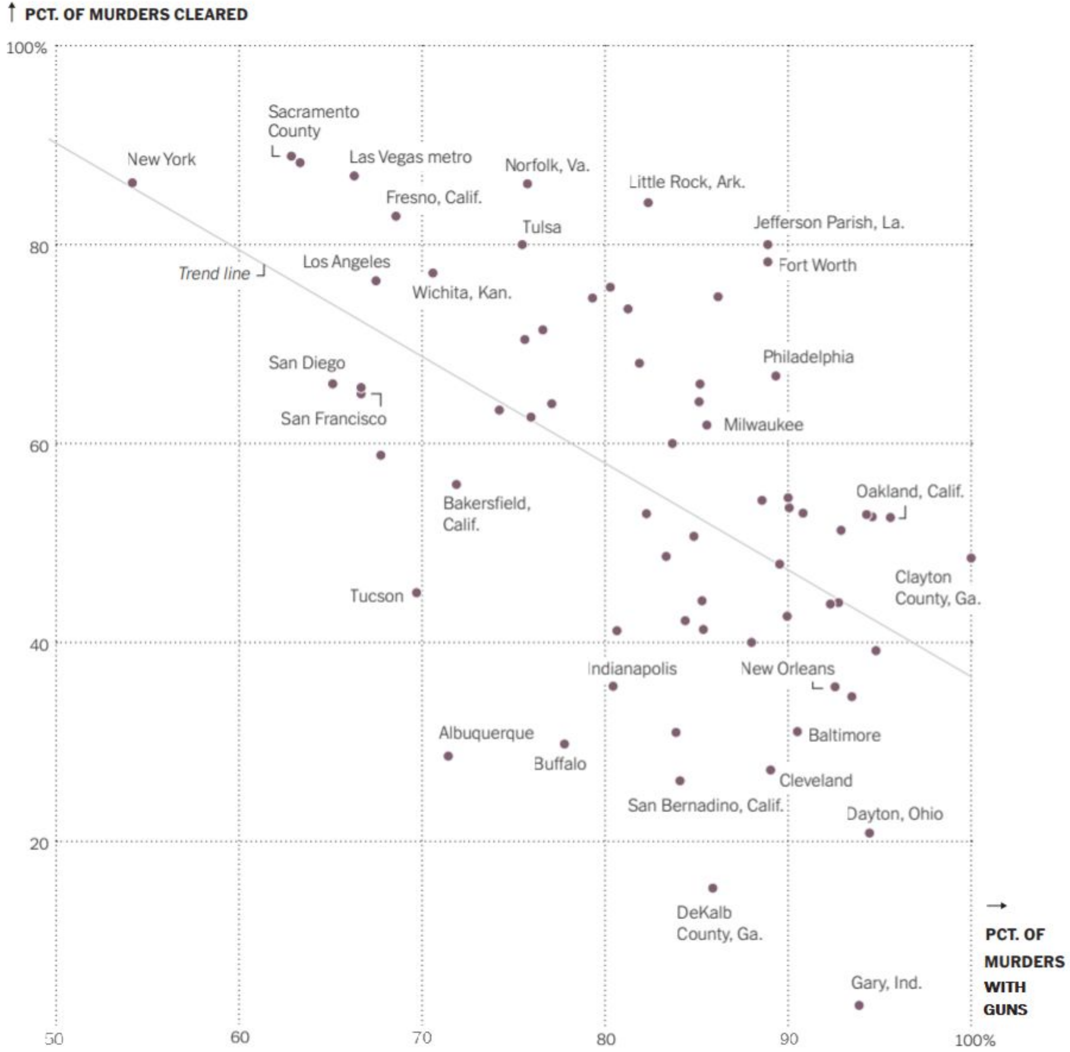
6. What **variables (for cases = *murders*)** were used to create this visualization? Be careful with wording. You should have at least 3 variables.
   a. Where the murder was committed
   b. If it involved a firearm
   c. If the murder was solved

7. Since each point on the graph represents a city, the cases here are actually “the cities.” What **variables are recorded about each *city*** in this visualization?
   a. How many murders?
   b. How many murders solved?
   c. How many murders committed w/a a firearm?



8. Can we draw the conclusion that the more murders there are committed using guns *causes* fewer murder cases to be solved? Discuss.
   a. Observational study so no, cannot draw a causal conclusion.

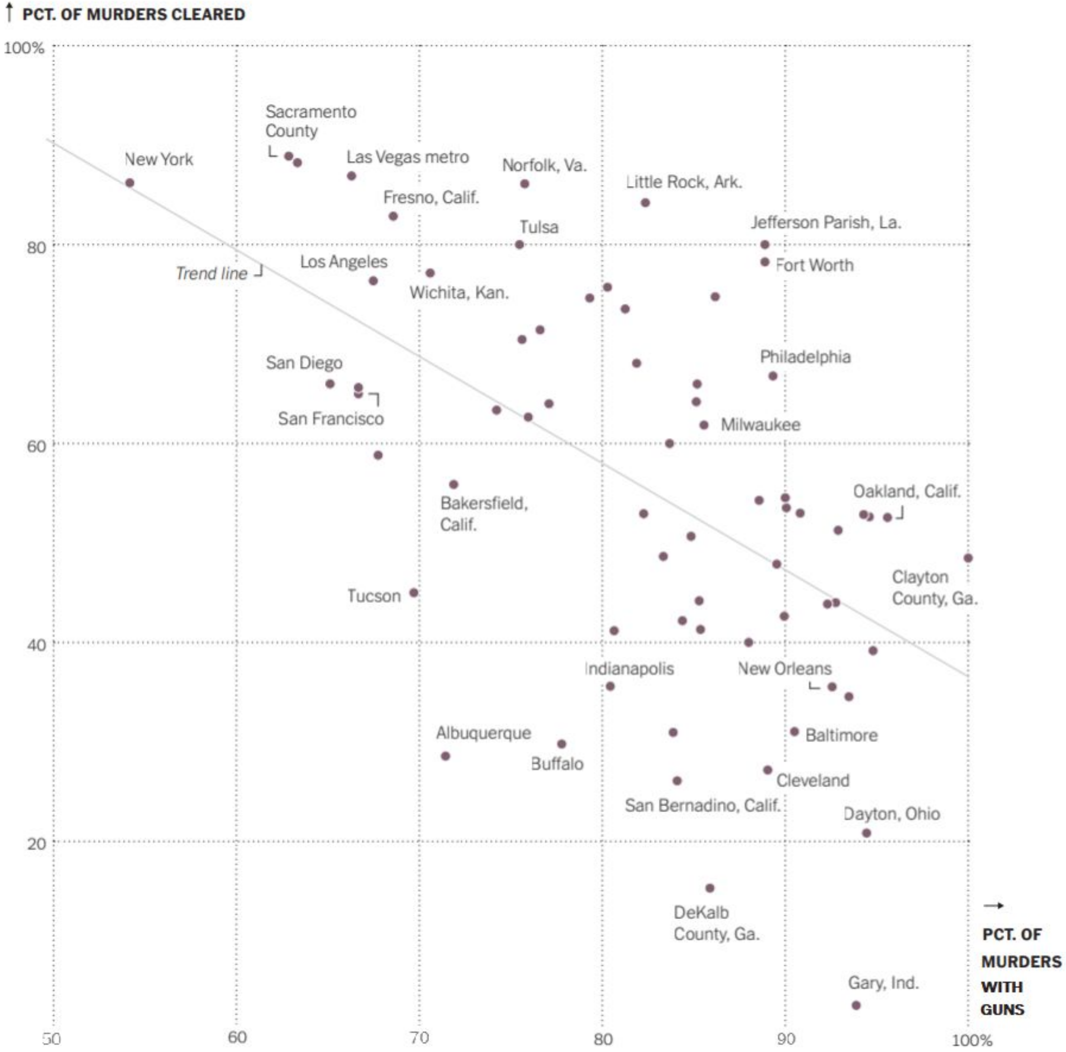
9. Can we generalize any conclusions we draw from this graph to all cities in the U.S.? Discuss.
   a. Yes. From this data the trend line indicates that if a murder was committed with a firearm it was less likely to be solved.



10. Propose a possible confounding variable that might explain the association between percentage of murders with guns and percentage of murders cleared.
    a. Police Force

11. Link between your confounding variable and the percentage of murders committed with guns:
    a. Dekalb County is significantly smaller than New York and most likely has a smaller police force and less regulations

12. Link between your confounding variable and the percentage of murders that are solved:
    a. More police = More solved murders

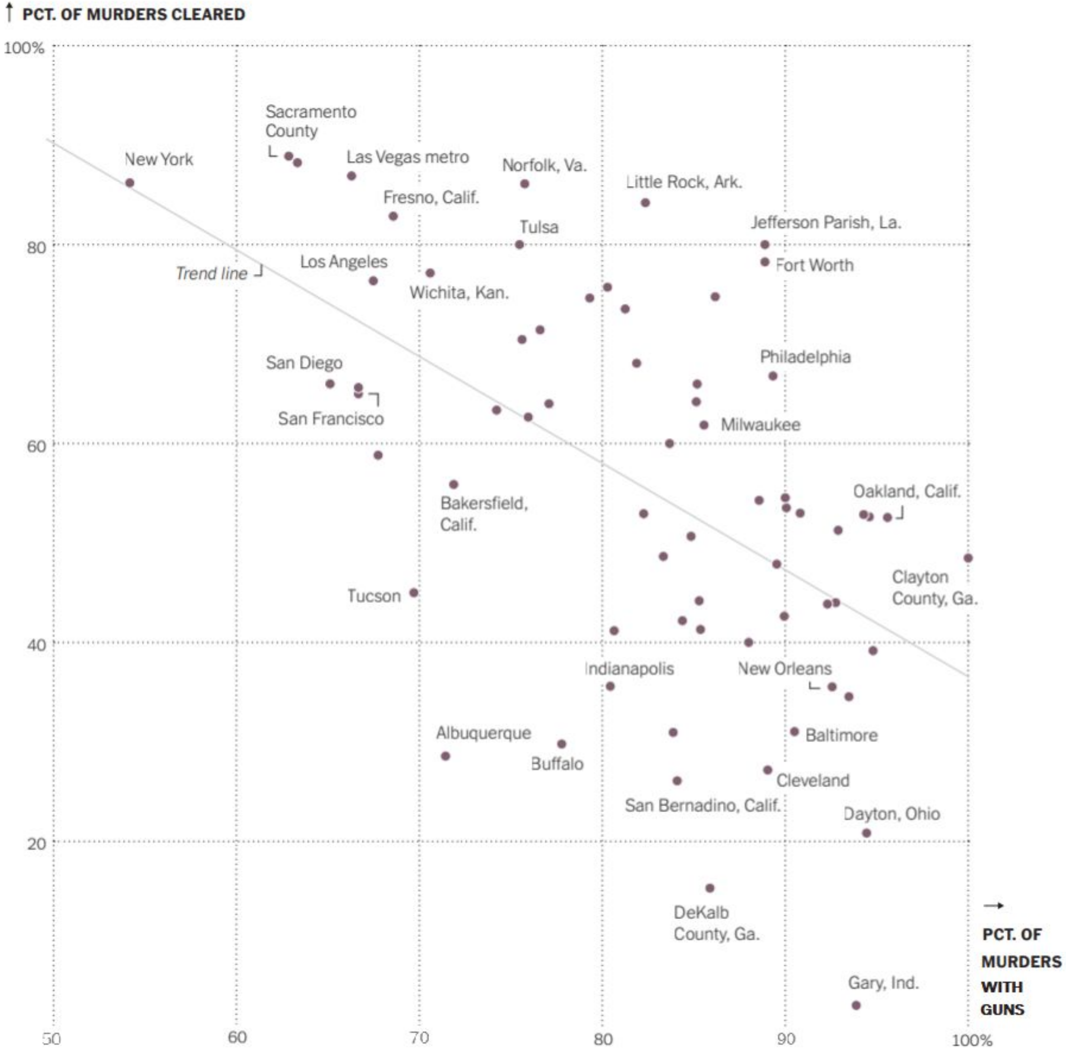


Group 10

This graph is from the NYTimes. Please don't search for the article just yet. Here is what the opening of the article says about the graph:

"Is the likelihood of solving a murder case less if the weapon is a gun?

This graph shows the relationship between the percentage of murders committed with guns for the 67 cities that had 30 or more murders in 2019 and the percentage of all murders solved."



. Tell me what you can about Norfolk, VA. Be specific.

orfolk had about 75% of their murders with guns ut they were able to clear 85% of their murders th guns

2. Tell me what you can about New York, NY. Be specific.

ew york has less than 60% murders of guns nd about more than 80% were solved.

3. Fill in the blank:

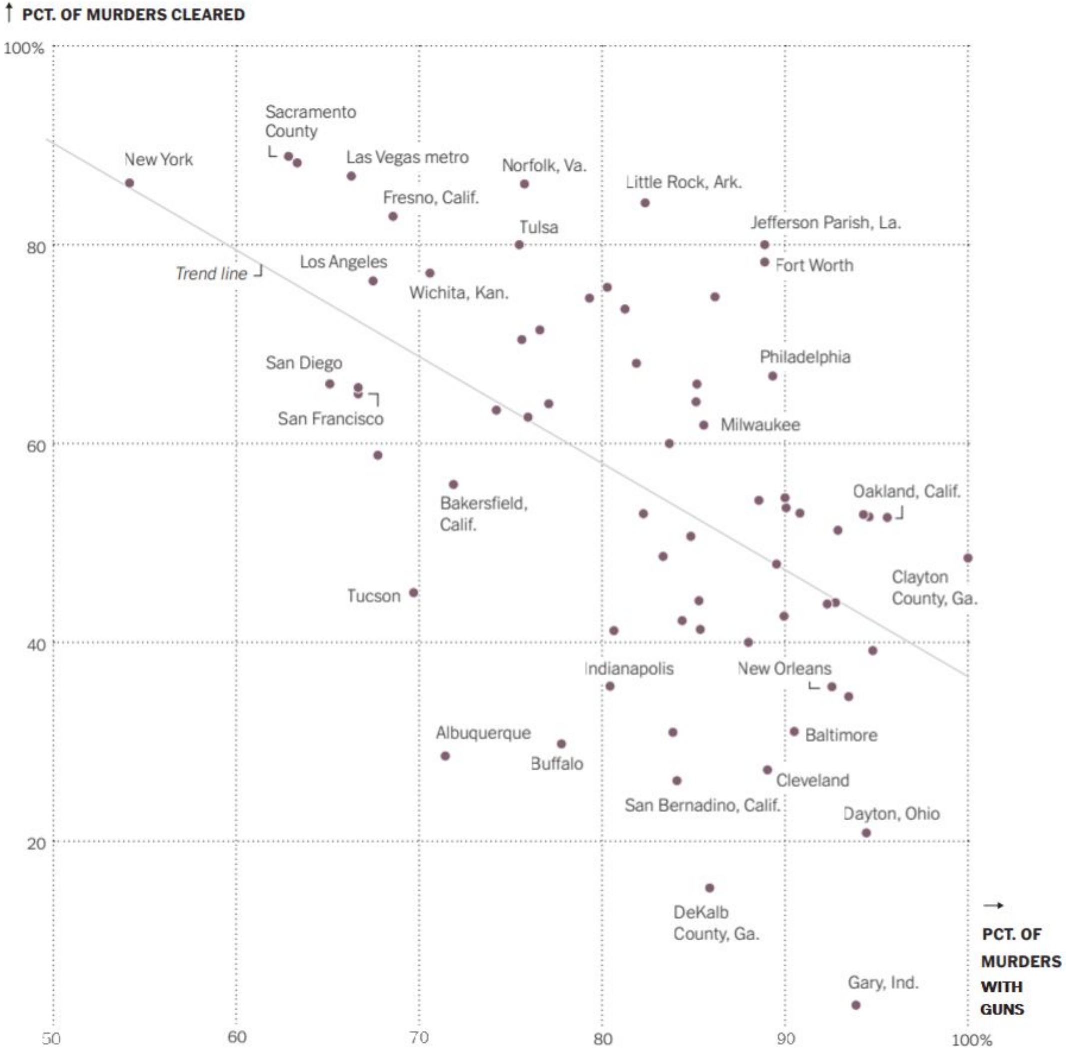
enerally speaking, for the cities represented in is graph, the greater the percentage of murders ommitted using a gun, the __Less__ the ercentage of murder cases that are solved.

4. What is the **sample** for this visualization?

he cities in the US that had over 30 or more urders with guns in 2019

5. What is the **population of interest** to the person who created this visualization, would you say?

The people who live in those cities



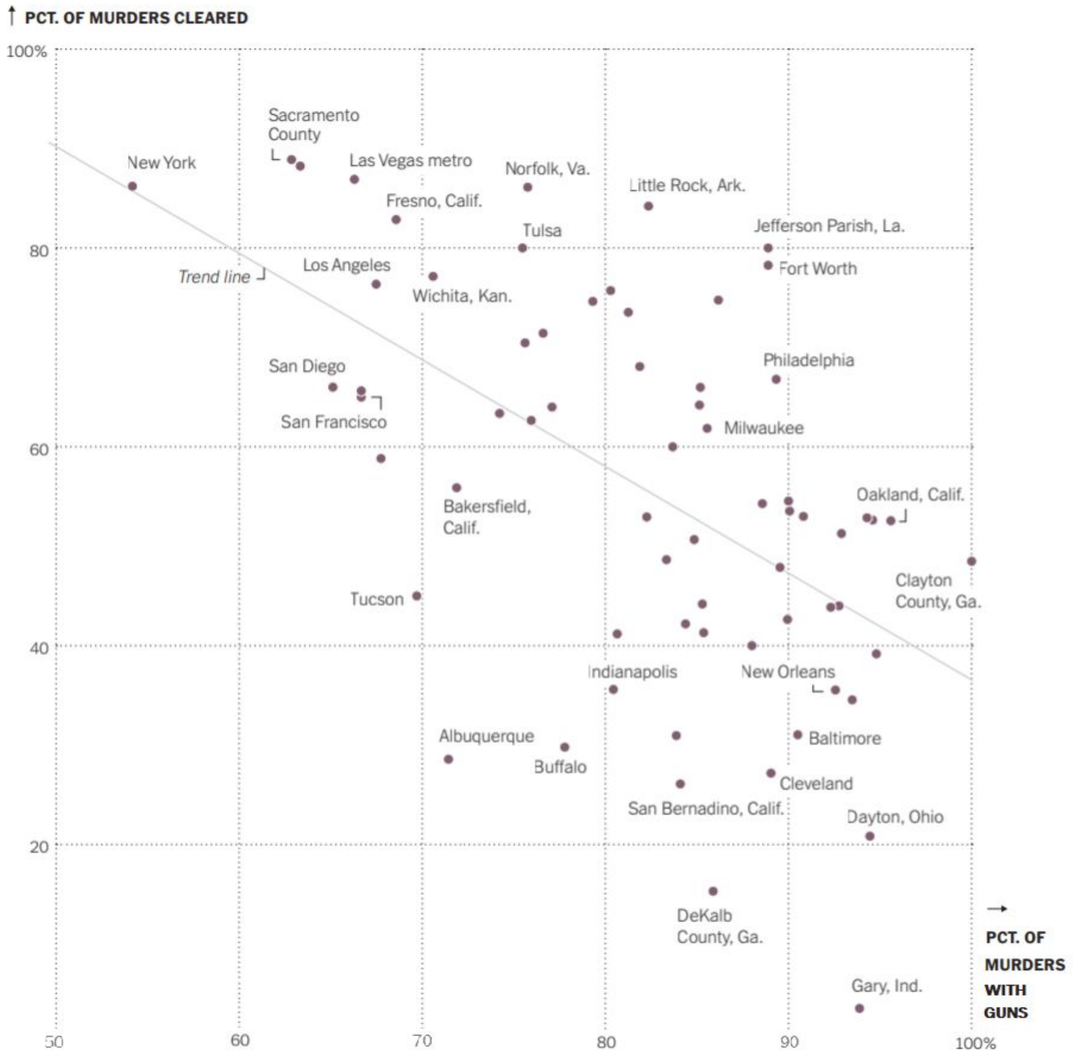
First, pretend the cases are "the murders".

6. What **variables (for cases = *murders*)** were used to create this visualization? Be careful with wording. You should have at least 3 variables.

Cities with 30 or more gun murders, pct. Of murders cleared, and pct. Of guns used in murders

7. Since each point on the graph represents a city, the cases here are actually "the cities." What **variables are recorded about each *city*** in this visualization?

Percentage of murders cleared and percentage of murders with guns

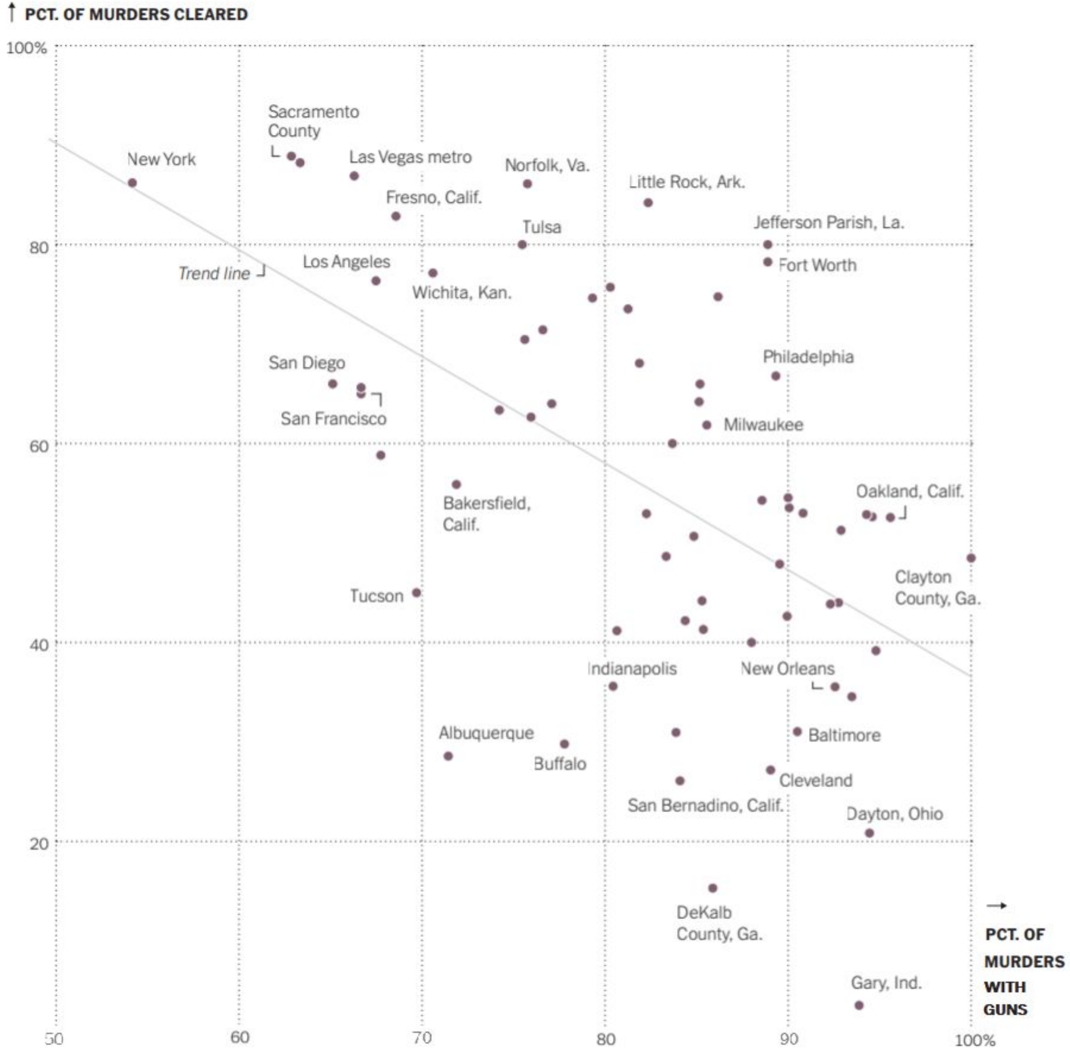


8. Can we draw the conclusion that the more murders there are committed using guns *causes* fewer murder cases to be solved? Discuss.

Can't conclude, observational study

9. Can we generalize any conclusions we draw from this graph to all cities in the U.S.? Discuss.

We cannot conclude this because the sample is not random and it doesnt correlate to all cities in the US



10. Propose a possible confounding variable that might explain the association between percentage of murders with guns and percentage of murders cleared.

The amount of murders with guns, some cities could have several more than another

11. Link between your confounding variable and the percentage of murders committed with guns:

12. Link between your confounding variable and the percentage of murders that are solved: